



Hybrid algorithms for enhanced efficiency and scalability of network-based tri-level interdiction models

Nafiseh Ghorbani-Renani¹ · Andrés D. González² · Kash Barker² 

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Abstract

This study focuses on optimizing resilience strategies for interdependent infrastructure networks (e.g., electric power, water supply, transportation system) against intelligent attacks. We formulate this problem as a tri-level defender-attacker-defender (DAD) model, which is known for its computational complexity. To address this challenge, we propose two novel hybrid decomposition-based algorithms: the hybrid Benders decomposition-based (HBD) algorithm and the hybrid set covering-based (HSC) algorithm. These algorithms efficiently solve the nested subproblems (master problem and subproblem) of the tri-level DAD formulation, incorporating metaheuristic algorithms for improved performance. The proposed algorithms are applied to a tri-level protection-interdiction-restoration model to optimize network resilience. Two case studies are used to evaluate the effectiveness of the solution techniques: (i) An interdependent system of water, gas, and power networks with 125 nodes and 164 links, and (ii) A simulated system of two networks with 252 nodes and 507 links. Our results demonstrate that both hybrid algorithms offer high-quality solutions with significantly improved computational efficiency compared to the existing exact solution method based on the set covering approach. In our comparison across different case studies and budget scenarios, we find that for higher budget scenarios, the HBD algorithm outperforms the HSC algorithm and is more computationally efficient. For lower budget scenarios, the performance of both algorithms is similar, with the HSC algorithm showing slightly better performance in terms of computational speed. Both algorithms show clear advantages over the set covering (CD) approach, particularly when available budget grows. This study highlights that the choice between the HBD and HSC algorithms depends on both the case study size and available budget, with the HBD algorithm being preferable in higher budget scenarios and the HSC algorithm being slightly more efficient in lower budget scenarios.

✉ Kash Barker
kashbarker@ou.edu

¹ Department of Systems and Enterprises, Stevens Institute of Technology, Hoboken, NJ, USA

² School of Industrial and Systems Engineering, University of Oklahoma, Norman, OK, USA

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1 Introduction

For several decades, interdiction models have received much attention due to their applicability in different domains, including defense and military applications (McMasters & Mustin 1970; Ghare et al. 1971; Patterson & Apostolakis 2007), controlling trafficking illicit materials (Pan et al. 2003; Morton et al. 2007; McLay et al. 2011; Baycik et al. 2018), facility location problems (Liberatore et al. 2011; Losada et al. 2012; Akbari-Jafarabadia et al. 2015; Akbari-Jafarabadi et al. 2017; Fathollahi Fard et al. 2018), and project management (Brown et al. 2009), among others. These models focus on understanding strategic decision-making processes where an entity (the attacker) seeks to disrupt a system, and another entity (the defender) attempts to neutralize or mitigate these disruptions.

A prominent framework used in modeling these strategic interactions is the Stackelberg game (1952), which involves two decision-makers in a hierarchical relationship, one as the leader and the other as the follower. While Stackelberg originally introduced this model in a competitive context, it has been particularly adapted to the attacker-defender (AD) model in the context of interdiction problems. In this adaptation, the attacker (the leader) makes the first move by selecting a target or disrupting a system, and the defender (the follower) responds by choosing the best strategy to counteract or mitigate the attacker's decision (Motto et al. 2005; Salmeron et al. 2004, 2009). The hierarchical nature of this interaction closely aligns with the decision-making structure in the Stackelberg game, making it a suitable framework for analyzing AD scenarios in network interdiction.

This study takes a step further by addressing the limitations of previous models through the introduction of a tri-level defender-attacker-defender (DAD) formulation, which adds an additional defense layer, making the problem more complex however providing more effective solutions for network protection. The tri-level structure reflects a more realistic sequence of decisions, where the attacker takes the first action, the defender responds, and an additional defense layer is introduced to further counteract the attack, thus improving network resilience. While DAD formulation significantly increases the complexity of solving these models, it provides more robust solutions for defending against strategic adversarial threats.

The motivation for using the Stackelberg game in this context stems from its ability to capture the sequential decision-making process between the attacker and defender. In our resilience-driven tri-level model (Ghorbani-Renani et al. 2020), the decision-makers at each level are as follows: (i) The attacker at the top level makes the first move to disrupt the system, (ii) The defender at the second level reacts to this disruption by implementing protection strategies, and (iii) The second defense layer at the third level reinforces the protection or repairs the system to ensure optimal resilience. This

tri-level structure mirrors the dynamics of a Stackelberg game, where each decision-maker's actions are strategically interdependent and influence the subsequent decisions of the other players.

The significance of studying this topic lies in its potential to enhance the resilience of critical networks against intentional attacks, a concern of growing importance in fields such as cybersecurity, infrastructure protection, and military strategy. As networks become more interdependent and vulnerabilities increase, it is crucial to develop optimization models that can effectively guide decision-making in response to adversarial threats. However, the complexity of solving such models, particularly with partial protection and interdiction actions, presents a significant challenge. Current solution methods face limitations in terms of computational efficiency and scalability, which emphasizes the importance of developing novel algorithms capable of handling these complex problems in a reasonable time.

In this study, we propose two hybrid decomposition-based algorithms to tackle the computational challenges inherent in solving the tri-level DAD models. These algorithms are designed to deliver high-quality solutions efficiently across a range of network interdiction scenarios, with a particular focus on DAD models involving partial protection and interdiction. The first algorithm, the *hybrid Benders decomposition-based* (HBD) algorithm, extends the traditional Benders decomposition method by incorporating a metaheuristic algorithm to handle a broad variety of tri-level formulations. The second, the *hybrid set covering-based* (HSC) algorithm, improves the efficiency of the covering decomposition (CD) approach by integrating a tailored metaheuristic algorithm. We demonstrate the effectiveness of these algorithms through two case studies, comparing their solution quality and computational efficiency against existing exact methods, CD approach used in Ghorbani-Renani et al. (2020).

The contributions of this study are twofold: (i) We present two novel hybrid decomposition-based algorithms with broad applicability to network interdiction models (e.g., AD and DAD formulations), offering high-quality solutions in a reasonable time frame, and (ii) We compare and analyze the performance of these algorithms for the resilience-driven DAD proposed by Ghorbani-Renani et al. (2020) under different network topologies and budget constraints with respect to an existing exact solution approach, the CD algorithm, providing insights into their practical utility for real-world applications.

The rest of the paper is structured as follows. Section 2 presents a detailed literature review on (i) The development of interdiction models and their application in network-based systems, and (ii) The primary solution methods to solve such models. Section 3 provides a general network interdiction formulation in which an interdictor seeks to maximize the cumulative of unmet demand across the system of interdependent networks by disrupting a set of links based on the budget availability. This problem is formulated as a tri-level DAD hierarchical formulation for interdependent network resilience. In Sect. 4, the hybrid solution techniques are presented with the general framework of their algorithms. In Sect. 5, we apply the proposed solution techniques to the tri-level optimization model developed by Ghorbani-Renani et al. (2020) for two systems of interdependent networks with different sizes and compare the solution time

and quality of the results with respect to the existing exact solution method. Finally, concluding remarks are provided in Sect. 6, with a discussion of future work.

2 Literature review

This paper is concerned with (i) Optimizing the resilience of interdependent infrastructure networks against adversaries using interdiction models, and (ii) Efficient solution approaches to solve the model in a reasonable time. Thus, this paper is related to two main areas of research: (i) The application of models of network protection-interdiction-restoration, and (ii) Existing solution algorithms for solving the proposed interdiction models. These two sectors will be discussed in the following subsections.

2.1 Interdiction models in network-based systems

Network-based systems broadly appear in various service, community, industrial, and economic environments. Infrastructure systems such as power, gas, and water supply are ubiquitous in community health and productivity. Thus, it is essential to have effective decision support tools to assist decision makers in ensuring the security and resilience of such systems under a variety of disruptions (e.g., natural disasters, malevolent attacks, operational failures) (Ma et al. 2013). Recently, Sharkey et al. (2020) conducted a thorough review in the resilience theory and its applications in network-based systems, with a focus on optimization techniques. Smith and Song (2020) reviewed the evolution of interdiction models, including their formulations and applications in real-world problems, with an emphasis on optimization techniques. In addition, they discussed primary solution approaches for dealing with interdiction models from dualization to reformulation to column-and-constraint generation approaches (Smith & Song 2020). In the following, we discuss interdiction models applied in common network-based systems.

2.1.1 Attacker-defender (AD) models

Decision makers should make prudent decisions about allocating finite resources (how much and where?) to harden the system in such a way to (i) Mitigate its vulnerability to disruption, and (ii) Minimize the long-term effect after the disruption, along with the decisions regarding the post-disruption restoration planning. Such a decision support tool can be effectively achieved using interdiction models. Interdiction models are formulated as a sequential game in the form of multi-level hierarchical framework with a corresponding hierarchy of decisions made at each level. Wollmer (1964) originally applied the interdiction model to a network flow problem by removing links in which the maximum flow between source and sink nodes is minimized. In a similar manner, shortest-path interdiction problems address situations in which the interdictor seeks to impair/lengthen a set of links such that the time or cost required to reach the sink node is maximized (Israeli & Wood 2002; Holzmann & Smith 2021; Nguyen & Smith 2021). Similarly, Azizi and Seifi (2023a, b) addressed a shortest

path network interdiction problem with incomplete information and multiple levels of interdiction intensity, considering uncertainty in the evader's perception of attack costs. Additionally, Punla-Green et al. (2024) considered an extension of the shortest path network interdiction problem, incorporating robustness to account for parameter uncertainty. In this work, they employed robust optimization techniques, where the interdicator accounts for uncertainties in the network parameters, providing a strategy that mitigates risks associated with worst-case scenarios in adversarial settings (Punla-Green et al. 2024). Interdiction models are considered as a powerful approach for pre-disruption strategies to reduce the vulnerability of network-based systems under intelligent attacks (Zhu et al. 2013). Nguyen et al. (2023) presented a novel fairness-aware AD model for optimal network operation and hardening. Sun and Sun (2023) introduced a novel two-level algorithm for distributed nonconvex optimization problems with complex constraints in network settings. A recent study conducted by Forghani et al. (2020) has used bi-level programming models to analyze partial attacks on capacitated nested facilities. In these models, the interdicator partially targets facilities, while the defender minimizes customer satisfaction costs by using available facilities or outsourcing. Hunt and Zhuang (2024) conducted a systematic literature review of AD games, focusing on modeling, solving, and protecting against strategic adversarial threats, while identifying challenges, assumptions, and future research directions.

A number of interdiction models have been developed for critical infrastructure networks, particularly power networks. Salmeron et al. (2004) initially assessed power grid vulnerability by developing a bi-level AD formulation. The proposed power grid interdiction model identifies critical system components, the outage of which would result in the maximum disruption to the system for a given attacker budget. In addition, they considered the cost of repair duration of damaged components in the objective function of an AD model for analyzing electric grid vulnerability.

2.1.2 Defender-attacker-defender (DAD) models

Brown et al. (2006) discussed that, although the AD model is applicable for vulnerability analysis of critical infrastructure networks by determining the set of critical components in the system, it does not necessarily determine the best (optimal) pre-disruption defensive strategy against an attacker's decision, which can mislead the decision maker to fortify the system inefficiently. Therefore, they introduced an additional level of defender action to the previous model, developing the tri-level DAD model for the power grid to enable the system to better withstand failure. In the proposed model, the first defender represents the system planner seeking to determine the best protection plan against the most destructive attacker decision. Subsequently, Yao et al. (2007), Alguacil et al. (2014), and Yuan et al. (2014) explored vulnerability mitigation of the electric power grid through a DAD model. In addition, interdiction models have received considerable attention for reducing the vulnerability of transportation networks under disruption. For example, Alderson et al. (2011) developed a DAD model for defending a transportation network against an intelligent adversary. Likewise, Alderson et al. (2014) and Sarhadi et al. (2015) applied this modeling framework to determine how to improve the operation of a rail network subject to disruption.

Wu et al. (2021) introduced a three-player game involving the attacker, defender, and operator of a water supply system, where the defender aims to minimize disruption impacts caused by the attacker while ensuring water demand is met. Hou et al. (2024) proposed a tri-level robust optimization framework to enhance distribution network resilience against extreme natural disasters like typhoons, integrating pre-disaster, in-disaster, and post-disaster strategies. The interdiction models discussed above have addressed a single-structured network. However, critical infrastructure systems may depend on each other to be operational, and thus a disruption in one network can spread to others (Sharkey et al. 2015).

Therefore, some researchers have been motivated to include interdependency among infrastructure systems in this modeling framework. For example, Ouyang (2017) proposed a tri-level DAD model to address the resilience of interdependent critical infrastructure systems under the worst spatially localized attacks. In this study, the operation of each infrastructure system is modeled by a network flow model with links connecting the interdependent systems. Ouyang (2017) quantifies the resilience of critical infrastructure systems immediately following a disruptive event and before any recovery efforts. There is substantial literature on optimal restoration of interdependent networks (González et al. 2016; Almoghatawi et al. 2019). However, a few researchers have incorporated restoration cost and/or recovery duration of disrupted components in interdiction models, particularly in DAD formulations. Ouyang and Fang (2017) considered the repair sequence of disrupted components with given recovery resources in a tri-level model. Ghorbani-Renani et al. (2020) proposed a protection-interdiction-restoration framework for a system of interdependent networks by considering work crew assignment and scheduling for restoring disrupted components. The model by Ghorbani-Renani et al. (2020) aims to optimize the restoration of a system of interdependent networks by minimizing the cumulative unmet demand over time by sequentially performing three levels: (i) Protection level to allocate resources (how much and where?) to harden the networks to simultaneously mitigate pre-disruption system vulnerability and enhance post-disruption system recoverability, (ii) Interdiction level to allocate resources to maximize the effects of the disruption, and (iii) Restoration level to minimize the long-term effect after the disruption through network component recovery. Similarly, Bao et al. (2019) developed a tri-level model to enhance the resilience of critical facilities under deliberate attacks. The proposed model considers the simultaneous decisions on fortifying the facilities before the disruption along with the restoration strategies of disrupted facilities following a disruptive incident. Although recovery of interdicted facilities has been incorporated to the model, the restoration decision has solely focused on the time required to repair a disrupted facility (e.g., the length of time in which the interdicted facility is not available to satisfy demand). Further, restoration decisions do not include work crew assignment and scheduling, and while the resilience of a facility may depend on the functionality of other facilities, Bao et al. (2019) do not incorporate the interdependency among facilities.

A recent study conducted by Parvasi et al. (2019) has also investigated the tri-level location problem of defense facilities, focusing on the r -interdiction median model over a finite time horizon, which aims to protect service facilities against severe attack scenarios and allow the system to use its maximum power to serve customers

after damage. The model is based on leader–follower games in the form of DAD, considering a time horizon to increase realism (Parvasi et al. 2019). Similarly, Alavi et al. (2024) presented a tri-level optimization model that integrates facility location and interdiction decisions, incorporating corrective redesign measures at the third level to improve system resilience after disruptions. The model, tested on a real supply system, demonstrates that redesign decisions significantly reduce disruption costs by reconfiguring the system post-interdiction.

2.2 Solution approaches for interdiction models

Generally, interdiction models are a multilevel optimization paradigm that formulates a hierarchical decision-making process between two or more opposing forces. This type of formulation is typically non-convex primarily due to the hierarchical structure of the problem, where decisions at each level are interdependent and affect the feasible region and objective function in complex ways. These problems often involve non-linear and integer variables that contribute to the non-convexity of the problem. The presence of multiple decision-makers or levels with interlinked decision variables leads to non-convex feasible regions and objective functions, making it challenging to guarantee global optimality (Martin 2007). Nabli et al. (2020) analyzed the computational complexity of sequential games formulated by multilevel problems and performed on a network application. Considering the widespread applications of interdiction models in different domains, the development of efficient solution methods has captured attention in recent years. Several solution approaches have been developed to deal with this type of models, generally classified as reformulation, decomposition, and heuristic-based methods. Below we discuss primary solution approaches along with their costs and benefits for solving such formulations.

Israeli and Wood (2002) introduced decomposition-based algorithms to solve the bi-level shortest path interdiction problem and applied valid inequalities to speed up the performance of the algorithms. The most common approach to reformulate the bi-level model as a single-level formulation is to employ the strong duality property of linear programs. For example, Motto et al. (2005) solved the bi-level AD model to optimality by such a reformulation. Salmeron et al. (2004) developed a decomposition-based heuristic algorithm to solve the bi-level interdiction model to find an acceptable interdiction plan from the perspective of an intelligent attacker. Likewise, Bier et al. (2007) proposed a heuristic method for dealing with the AD model for assessing power transmission system vulnerability based on a greedy algorithm such that at each iteration, the transmission line with the maximum load is interdicted. Though this approach does not provide an optimal solution, it is an inexpensive and practical method for determining promising interdiction plans in relatively complex systems. Azizi and Seifi (2023a, b) proposed algorithms for solving symmetric and asymmetric models of shortest path network interdiction, focusing on efficiency improvements and overcoming challenges in near-optimal solutions for the asymmetric model. The proposed methods significantly outperform traditional duality-based approaches, particularly for large instances and those with substantial interdiction budgets. Baycik (2022) applied

machine learning-based approaches, such as decision trees and random forest regression, to solve the maximum flow network interdiction problem. The results show that machine learning based approaches significantly improve computational efficiency, especially for large instances, compared to traditional integer programming methods (Baycik 2022).

Yao et al. (2007) applied the CD method to solve the tri-level power grid interdiction formulation. This approach, which is an extension of a set covering decomposition method (Israeli & Wood 2002), iteratively solves nested bi-level problems to reach the optimal solution. However, this decomposition-based solution method is based upon the explicit enumeration approach which requires a substantial amount of time to implement and solve even for the small-sized networks (Alguacil et al. 2014). Ghorbani-Renani et al. (2020) introduced an improvement to the CD algorithm to solve the tri-level DAD model. Although this development increases the efficiency of the algorithm, it still suffers from the computational burden imposed by large-size problems. Yuan et al. (2014) developed the column-and-constraint generation (C&CG) method to solve the tri-level power network model to optimality, outperforming the CD algorithm with respect to computational effort. Note that Zeng and Zhao (2013) originally presented the C&CG algorithm to derive optimal solutions for two-stage robust optimization problems. Alguacil et al. (2014) proposed a two-stage solution strategy to find the optimal solution for the DAD problem by converting it to its equivalent bi-level model through dualizing the third level and merging it with the second level. Then, the transformed model was solved to optimality by an implicit enumeration algorithm. While the proposed two-stage solution method is computationally more efficient than CD algorithm, it may not be applicable for multiple attacks and protection decisions (Yuan et al. 2014). On the other hand, the two solution approaches proposed by Yuan et al. (2014) and Alguacil et al. (2014) are based on converting the two levels into a single level by taking the dual of the inner level. Such a conversion can only be applicable for linear programming problems in the lower level since Karush–Kuhn–Tucker optimality conditions are not satisfied for either nonlinear or mixed integer problems (Israeli & Wood 2002; Wood 1993, 2011). In addition, even for the case that a strong dual formulation can be found, the subsequent interdiction model is usually transformed to a large linear mixed-integer formulation, which is computationally expensive to solve (Lozano & Smith 2016).

Cappanera and Scaparra (2011) applied an implicit enumeration algorithm to solve the tri-level shortest path interdiction problem. They enhanced the algorithm by utilizing a heuristic approach to generate valid inequalities, leading to a reduction in the search tree size. Ouyang (2017) solved exactly the proposed tri-level DAD model by iteratively solving nested models constructed from the original model. The devised algorithm is based on the decomposition algorithm developed by Alderson et al. (2011, 2014) and incorporates the concept of “maximal component set” and Benders decomposition. However, Ouyang (2017) discussed that utilizing some heuristic algorithms (e.g., genetic algorithm) as alternatives may improve the efficiency of the algorithm. Ouyang and Fang (2017) utilized a decomposition algorithm to solve the developed DAD mathematical framework to optimality. Although the proposed algorithm is an exact solution method, it is applied to a very small illustrative example of nine nodes.

However, infrastructure networks are often large-sized problem, and due to the general features of DAD models, the computational complexity of the algorithm grows exponentially in such an application. Note that along with other factors (e.g., complexity of the formulation by itself), the run time of the DAD model is highly sensitive to the amount of budget available for the attacker and protector, since increases in budget result in more possible combinations for interdiction and protection (e.g., the feasibility space is expanded).

From the above discussion, it can be concluded that exact solution methods for solving interdiction models typically have limitations from modeling and computational complexity perspectives. Especially, interdiction models with a nonconvex inner level remain challenging to solve and only particular cases of this type of model have been explored recently. In addition, the computational complexity of DAD models emerges when we are dealing with a large-sized problem with higher budget availabilities. In this regard, hybrid algorithms incorporating exact solution algorithm and heuristic method can increase the efficiency of the solution approach (Akbari-Jafarabadi et al. 2017). Aksen and Aras (2012) solved a bi-level facility location problem with Tabu search. Similarly, Sadati et al. (2020) proposed an AD model for identifying the most critical depots in a vehicle routing context and used Tabu search to solve the lower level. Stozhkov et al. (2017) considered a bi-level extension of the assignment problem inspired by the network interdiction model and solved it with a greedy-based approach. Mahmoodjanloo et al. (2016) solved a DAD model for facility protection with (i) An exact solution approach based on an implicit enumeration algorithm and (ii) A heuristic-based method using a genetic algorithm. Bao et al. (2019) proposed a tri-level DAD model for enhancing the resilience of facilities under intelligent attacks and applied the ant colony system algorithm. However, the results discussed in this study have only focused on (i) Solving the model under variations of model parameters (e.g., budget level, numbers of customers and facilities), and (ii) The advantages of the tri-level formulation relative to the bi-level model. There is also no discussion on efficiency of the solution approach to deal with the complexity of the formulation and comparing it with other existing methods (e.g., in terms of computation speed and quality of the solution, among others).

Table 1 summarizes the key papers discussed, outlining their contributions to the field of interdiction models in network-based systems. It also provides a valuable comparison with this paper, highlighting the unique contributions of our study in relation to existing research articles. Several contributions have been made to explore hybrid algorithms that are candidates for solving DAD models in reasonable time. However, they could not entirely handle a wide variety of DAD models with different modeling features (e.g., DAD models with partial or total fortification and interdiction, DAD models with a nonconvex inner level). In addition to the solution speed, the solution quality is a concern in applying heuristic solution approaches to this type of formulation. Therefore, we are motivated to devise hybrid decomposition-based algorithms to effectively deal with a wide range of network interdiction models, particularly DAD models with different modeling characteristics and application sizes, while not sacrificing the quality of the provided solutions. Unlike previous studies that focus on isolated networks, our approach integrates interdependencies among critical infrastructure systems and incorporates a recovery process in the third level of the

Table 1 Comparative analysis of key studies on interdiction models in network-based systems and their contributions

References	Model type	Application domain	Interdependent networks	Recovery process	Solution approach
Bier et al. (2007)	AD	Electric power grid	✗	✗	Greedy algorithm
Motto et al. (2005)	AD	Electric power grid	✗	✗	Reformulation
Salmeron et al. (2004)	AD	Electric power grid	✗	✓	Decomposition-based heuristic algorithm
Forghani et al. (2020)	AD	Service facilities	✗	✗	Heuristic algorithm
Aksen and Aras (2012)	AD	Service facilities	✗	✗	Tabu search
Sadati et al. (2020)	AD	Vehicle routing problem	✗	✗	Tabu search
Nguyen et al. (2023)	AD	Distributed network infrastructure	✗	✗	Reformulation
Stozhkov et al. (2017)	AD	Resource allocation networks	✗	✗	Greedy-based approach
Baycik (2022)	AD	Infrastructure networks	✗	✗	Machine learning-based approaches
Brown et al. (2006)	DAD	Electric power grid	✗	✗	Benders decomposition
Yao et al. (2007)	DAD	Electric power grid	✗	✗	Covering decomposition
Alguacil et al. (2014)	DAD	Electric power grid	✗	✗	Two-stage solution strategy
Yuan et al. (2014)	DAD	Electric power grid	✗	✗	Column-and-constraint generation
Alderson et al. (2011)	DAD	Transportation networks	✗	✗	Decomposition algorithm
Alderson et al. (2014)	DAD	Rail network	✗	✗	Decomposition algorithm
Sarhadi et al. (2015)	DAD	Rail network	✗	✗	Heuristic algorithm
Wu et al. (2021)	DAD	Water supply network	✗	✗	Heuristic algorithm
Hou et al. (2024)	DAD	Distribution network	✗	✓	Column-and-constraint generation

Table 1 (continued)

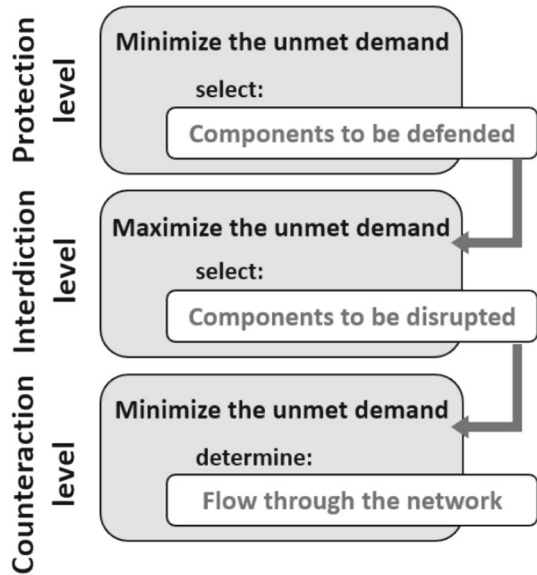
References	Model type	Application domain	Interdependent networks	Recovery process	Solution approach
Ouyang (2017)	DAD	Critical infrastructure networks	✓	✗	Decomposition algorithm
Ouyang and Fang (2017)	DAD	Critical infrastructure networks	✗	✓	Decomposition algorithm
Ghorbani-Renani et al. (2020)	DAD	Critical infrastructure systems	✓	✓	Covering decomposition
Bao et al. (2019)	DAD	Service facilities	✗	✓	Ant colony algorithm
Parvasi et al. (2019)	DAD	Service facilities	✗	✗	Heuristic algorithm
Alavi et al. (2024)	DAD	Service facilities	✗	✗	Column-and-constraint generation
Cappanera and Scaparra (2011)	DAD	Transportation networks	✗	✗	Heuristic algorithm
Mahmoodjanloo et al. (2016)	DAD	Service facilities		✗	Heuristic algorithm
This study	DAD	Critical infrastructure networks	✓	✓	Hybrid decomposition-based algorithm

model (subsequent defender level). This dual focus on both fortification and restoration enhances the resilience of the network, setting our work apart from others. We propose two hybrid solution approaches: one that uniquely extends Benders decomposition with a tailored metaheuristic algorithm, and another that efficiently modifies the CD approach by integrating it with a tailored metaheuristic. The proposed hybrid algorithms are shown to be effective, delivering high-quality solutions in reasonable time compared to the existing exact but sophisticated CD approach.

3 Problem description

This study is concerned with the optimization of resilience strategies for interdependent infrastructure networks (e.g., electric power, water supply) under intelligent attacks. We model this problem as a tri-level DAD hierarchical formulation with a corresponding hierarchy of decisions. This model includes a series of nested optimization formulations controlled by a set of constraints and decision variables in which the decision made at the top level affects the decisions of other levels. Figure 1 is a general depiction of the three levels involved in this problem: (i) Protection, (ii) Interdiction, and (iii) Counteraction. The protection level implements strategies for

Fig. 1 Tri-level DAD process



hardening network components such as increasing the capacity of the network component, enabling backup capabilities, and burying specific components in the case of electric power networks (Alderson et al. 2011; Ouyang & Fang 2017). The interdiction level identifies the set of components that would, if disrupted, result in the maximum damage through the system for a given attack budget. This level does not necessarily require a human “attacker” and could represent a worst-case natural disaster (note that terms related to the disruptive agent such as attacker and interdictor are used interchangeably through this paper). The disruption (interdiction) can take the form of a reduction in the capacity of the network components (either nodes or links). Finally, the counteraction level reacts after disruption perceived from the attacker with the aim of minimizing the system loss. The counteraction level can be performed either (i) By solving a generic and simplified optimal network flow problem to minimize the system loss (Alguacil et al. 2014; Yuan et al. 2014), or in a more comprehensive manner (ii) By solving an optimal network flow problem accounting for restoration process of disrupted components to minimize the long-term effect after the disruption through the network component recovery (Ouyang 2017; Ghorbani-Renani et al. 2020). To properly demonstrate various steps of the proposed hybrid solution algorithms, we first propose a general framework for the tri-level DAD model in which the recovery process has not been considered through the counteraction level. However, for experimental purposes, we utilize a more comprehensive model accounting for the restoration process of disrupted network components through the counteraction level as recourse decisions in the planning of network fortification, and we discuss the results. The comprehensive model, the tri-level DAD model that includes restoration, has been presented in detail in the appendix (Ghorbani-Renani et al. 2020).

3.1 Mathematical formulation

In this problem, we focus on the system of interdependent networks defined by a set of nodes connected with a set of links where the service provided by each network is represented by flow through it. We assume that each network is responsible for providing a specific type of service (commodity) across the network by satisfying the demand nodes of the network (e.g., the gas network provides gas service, the water supply network provides water requirements). We consider a single commodity flowing through a network, but it can be easily extended to account for multiple commodities. We consider physical interdependency for networks where the functionality of a set of nodes in one or more networks enable the functionality of a node in another network through interdependency links. In this problem, an intelligent attacker, who has complete information about the network, intentionally interdicts the system. The protector, on the other hand, defends (fortifies) the system in response to such prospective interdiction. There are known budgets available for protecting and for interdicting the components of the system of networks. There is a known fixed cost associated with protecting or interdicting each network component. Although infrastructure networks have different features that are guided by different physical principles, we approximated that infrastructure networks follow the general network flow problem. In the basic formulation in Eqs. (1)–(9), we assume that links are the only component that can be protected and interdicted, however the model can be easily extended to account for nodes as well. In this problem, we consider a set K of interdependent infrastructure networks. Each network $k \in K$ is represented by $G^k = (N^k, A^k)$, where:

- N^k is the set of nodes in network k , indexed by $i \in N^k$,
- A^k is the set of links between nodes in network k , indexed by $(i, j) \in A^k$.

The nodes in each network are categorized as:

- Supply nodes $N_+^k \subseteq N^k$
- Demand nodes $N_-^k \subseteq N^k$
- Transshipment nodes $N_=^k \subseteq N^k \setminus \{N_+^k, N_-^k\}$

The amount of supply and demand at node $i \in N_+^k$ and $i \in N_-^k$ in network k are denoted as s_i^k and d_i^k , respectively. Links in the network can either be protected or interdicted, and flow through these links is subject to the functionality status of the links.

We model the physical interdependency between different networks. Specifically, if (i, k) in network k physically depends on node $(\bar{i}, \bar{k})$ in network $\bar{k}$, then a positive flow through the link in network k depends on the operational status of the corresponding parent node in network $\bar{k}$. The set of interdependent nodes between networks is denoted as Ψ , where: $((i, k), (\bar{i}, \bar{k})) \in \Psi$ if node $i \in N^k$ physically depends on $\bar{i} \in N^{\bar{k}}$, where $N^k \cap N^{\bar{k}} = \emptyset$, $A^k \cap A^{\bar{k}} = \emptyset$ and $\forall k, \bar{k} \in K : k \neq \bar{k}$

3.1.1 Decision variables and model structure

The model is a tri-level optimization problem, where decisions are made by three different decision-makers: (i) The first defender decides which links to protect, (ii) The attacker decides which links to interdict, and (iii) The subsequent defender reacts to the attacker's decisions by solving the network flow problem, ensuring that the flow is optimized in the remaining operational network. The decision variables at each level are defined as follows:

- i. First defender's decisions (Level 1):
 - $y_{ij}^k \in \{0,1\}$: binary variable indicating whether link (i, j) in network k is protected by the first defender.
 - The protection cost for link (i, j) is CP_{ij}^k and the total budget for the first defender is BP .
- ii. Attacker's decisions (Level 2):
 - $z_{ij}^k \in \{0,1\}$: binary variable indicating whether link (i, j) in network k is interdicted by the attacker.
 - The interdiction cost for link (i, j) is CI_{ij}^k , and the total budget for the attacker is BI .
- iii. Subsequent defender's decisions (Level 3):
 - $f_{ij}^k \in \{0,1\}$: binary decision variables indicating whether link (i, j) in network K is functional (i.e., operational) after the first defender's protection and the attacker's interdiction.
 - x_{ij}^k : continuous decision variables providing the flow through link (i, j) in network K .
 - m_i^k : continuous decision variables denoting actual demand being met at node $i \in N_-^k$.

3.1.2 Objective function

The tri-level DAD model aims to deliver the cumulative weighted fraction of unmet demand, such that this value is minimized by the first defender, maximized by the attacker, and minimized by the subsequent defender. The objective function is shown in Eq. (1).

$$\mathbb{Q} = \min_y \max_z \min_{\{f, x, m\}} \left[\frac{\sum_{k \in K} \sum_{i \in N_-^k} w_i^k (d_i^k - m_i^k)}{\sum_{k \in K} \sum_{i \in N_-^k} w_i^k d_i^k} \right] \quad (1)$$

where

Parameters w_i^k represent the weight of demand node $i \in N_-^k$, emphasizing the importance of the node which can be adjusted based on different aspects including locations of the demand nodes (e.g., near hospitals, shelters, populated or more vulnerable areas).

3.1.3 Constraints

1. Budget constraints

- For the first defender: the total protection cost for the first defender should not exceed the available budget as shown in Eq. (2).

$$\sum_{k \in K} \sum_{(i, j) \in A^k} C P_{ij}^k y_{ij}^k \leq B P \quad (2)$$

- For the attacker: the total interdiction cost for the attacker should not exceed the available budget as shown in Eq. (3).

$$\sum_{k \in K} \sum_{(i, j) \in A^k} C I_{ij}^k z_{ij}^k \leq B I \quad (3)$$

2. Link functionality status (Subsequent defender):

- A link (i, j) is functional if it is either not interdicted or if it is protected as shown in Eq. (4).

$$1 + y_{ij}^k - z_{ij}^k \geq f_{ij}^k \quad \forall (i, j) \in A^k, \forall k \in K \quad (4)$$

3. Flow conservation (Subsequent defender):

- For each node i in network k , the flow balance should be maintained as shown in Eq. (5).

$$\sum_{(i, j) \in A^k} x_{ij}^k - \sum_{(j, i) \in A^k} x_{ji}^k \begin{cases} \leq s_i^k & \forall i \in N_+^k, \forall k \in K \\ = 0 & \forall i \in N_-^k, \forall k \in K \\ \leq -m_i^k & \forall i \in N_-^k, \forall k \in K \end{cases} \quad (5)$$

4. Demand node constraint (Subsequent defender):

- The amount of demand met at each demand node $i \in N_-^k$ cannot exceed the demand required at that node, ensuring that demand is satisfied to an extent no greater than the actual demand as shown in Eq. (6).

$$m_i^k \leq d_i^k \forall i \in N_-^k, \quad \forall k \in K, \quad (6)$$

5. Flow through links (Subsequent defender):

- The flow through each link is constrained by its capacity and the functionality status of the link as shown in Eq. (7).

$$x_{ij}^k \leq u_{ij}^k f_{ij}^k \forall (i, j) \in A^k, \quad \forall k \in K \quad (7)$$

6. Physical interdependency constraints (Subsequent defender):

- The flow on a link is only possible if the corresponding parent node (in another network) is functional as shown in Eq. (8). This ensures that interdependent links in different networks function only if the interdependent nodes are operational.

$$x_{ij}^k \leq u_{ij}^k f_{\bar{i}\bar{j}}^{\bar{k}} \quad \forall (i, j) \in A^k, \forall k, \bar{k} \in K, \forall (\bar{i}, \bar{j}) \in A^{\bar{k}} | \\ ((i, k), (\bar{j}, \bar{k})) \in \Psi \text{ or } ((j, k), (\bar{j}, \bar{k})) \in \Psi \quad (8)$$

7. Binary and non-negative decision variables:

- The decision variables are binary or continuous as follows:

$$y_{ij}^k \in \{0, 1\}, z_{ij}^k \in \{0, 1\}, f_{ij}^k \in \{0, 1\}, x_{ij}^k \geq 0, m_i^k \geq 0 \quad \forall (i, j) \in A^k, \forall i \in N_-^k, \forall k \in K \quad (9)$$

4 Hybrid solution techniques

The two solution methods are decomposition-based in which, by iteratively solving the master problem and subproblem, the final solution for the tri-level DAD model is obtained. We detail the proposed solution algorithms in the following subsections. Note again that for the purpose of presenting the proposed solution approaches, we first proposed a general and simplified framework for the tri-level DAD model in which the recovery process has not been considered through the counteraction level. However, the two solution methods that will be discussed subsequently can be easily applied to other types of interdiction models with the same procedure presented in this paper.

4.1 Hybrid benders decomposition-based (HBD) algorithm

The HBD algorithm is a novel extension of Benders decomposition with a metaheuristic algorithm tailored to efficiently address network-based interdiction formulations with different model characteristics. Under the fixed attacker decision variables ($\widehat{Z}$) from every algorithm iteration, the tri-level DAD model is transformed to the bi-level min-min formulation. Equations (10)–(18) show the master problem of the tri-level formulation including a subset of possible attack plans (Yuan et al. 2014). To develop the master problem, we assume a set of attack plans $\widehat{Z} = \{\widehat{Z}^1, \dots, \widehat{Z}^C\} \subseteq Z$ that are indexed by the iteration $c = 0, 1, \dots, C$, and we define a set of decision variables f_{ij}^{kc} , x_{ij}^{kc} , and m_i^{kc} for the associated attack plan for the given iteration. By defining a new set

of decision variables and constraints, the master problem is constructed based on the Benders framework (Zeng & Zhao 2013; Yuan et al. 2014) as shown in Eqs. (10)–(18). Note that the master problem is updated at each iteration and the new set of attacker decisions is generated with the current iteration index. Subsequently, the new set of decision variables f_{ij}^{kc} , x_{ij}^{kc} , and m_i^{kc} and constraints are added to the formulation for the corresponding attacker decision at the given iteration. Note that the first defender decision variables remain unchanged as shown in constraints (13) and (18). Solving the optimization model in Eqs. (10)–(18) provides the best (optimal) solution for the first defender decision variables, which will be the input to the subproblem.

$$\min \delta \quad (10)$$

Subject to :

$$\delta \geq \left[\frac{\sum_{k \in K} \sum_{i \in N_-^k} w_i^k (d_i^k - m_i^{kc})}{\sum_{k \in K} \sum_{i \in N_-^k} w_i^k d_i^k} \right] \quad \forall c = 0, 1, \dots, C \quad (11)$$

$$\sum_{k \in K} \sum_{(i, j) \in A^k} CP_{ij}^k y_{ij}^k \leq BP \quad (12)$$

$$1 + y_{ij}^k - \hat{z}_{ij}^{kc} \geq f_{ij}^{kc} \quad \forall (i, j) \in A^k, \forall k \in K, \forall c = 0, 1, \dots, C \quad (13)$$

$$\sum_{(i, j) \in A^k} x_{ij}^{kc} - \sum_{(j, i) \in A^k} x_{ji}^{kc} \begin{cases} \leq s_i^k & \forall i \in N_+^k, \forall k \in K, \forall c = 0, 1, \dots, C \\ = 0 & \forall i \in N_-^k, \forall k \in K, \forall c = 0, 1, \dots, C \\ \leq -m_i^{kc} & \forall i \in N_-^k, \forall k \in K, \forall c = 0, 1, \dots, C \end{cases} \quad (14)$$

$$m_i^{kc} \leq d_i^k \quad \forall i \in N_-^k, \forall k \in K, \forall c = 0, 1, \dots, C \quad (15)$$

$$x_{ij}^{kc} \leq u_{ij}^k f_{ij}^{kc} \quad \forall (i, j) \in A^k, \forall k \in K, \forall c = 0, 1, \dots, C \quad (16)$$

$$x_{ij}^{kc} \leq u_{ij}^k f_{\bar{j}}^{\bar{k}c} \quad \forall (i, j) \in A^k, \forall k, \bar{k} \in K, \forall (\bar{i}, \bar{j}) \in A^{\bar{k}} | \\ ((i, k), (\bar{j}, \bar{k})) \in \Psi \text{ or } ((j, k), (\bar{j}, \bar{k})) \in \Psi, \forall c = 0, 1, \dots, C \quad (17)$$

$$y_{ij}^k \in \{0, 1\}, f_{ij}^{kc} \in \{0, 1\}, x_{ij}^{kc} \geq 0, m_i^{kc} \geq 0 \quad \forall (i, j) \\ \in A^k, \forall i \in N_-^k, \forall k \in K, \forall c = 0, 1, \dots, C \quad (18)$$

By solving the master problem, the protection plan is determined, which will be the input of the subproblem. And by fixing the protector decision variables ($\hat{Y}$) found through the master problem, the subproblem is constructed as a bi-level max–min formulation including attacker and subsequent defender levels. By solving the subproblem, the interdiction plan is determined and serves as the input of the master problem. To solve the bi-level max–min formulation through the subproblem, we utilize a metaheuristic algorithm for dealing with a wide variety of bi-level formulations in an efficient manner. Metaheuristic algorithms are designed to address various

kinds of complex models and have received considerable attention for solving bi-level programs (Randall & Abramson 2001). For example, the genetic algorithm (Holland 1992) is a metaheuristic approach that has been successfully employed in solving bi-level formulations (Wang et al. 2007; Calvete et al. 2008; Hecheng & Yuping 2008; Arroyo & Fernandez 2009). In addition, this method can be applied to models with either integer or continuous decision variables in the first level. The particle swarm optimization algorithm (Kennedy & Eberhart 1995) is another metaheuristic approach that has been widely applied to bi-level formulations (Gao et al. 2011; Marinakis & Marinaki 2013). There are other kinds of metaheuristics that can be efficiently utilized in solving bi-level programs (e.g., simulated annealing, ant colony optimization). Talbi (2013) offers a review of the application of metaheuristics for solving ‘programming problems. The main advantage of metaheuristics is their applicability to a bi-level program regardless of its complexity in the inner level (e.g., a bi-level formulation with a nonconvex inner level). Figure 2 illustrates the general structure of the proposed tailored metaheuristic algorithm applied to solve the subproblem. The proposed

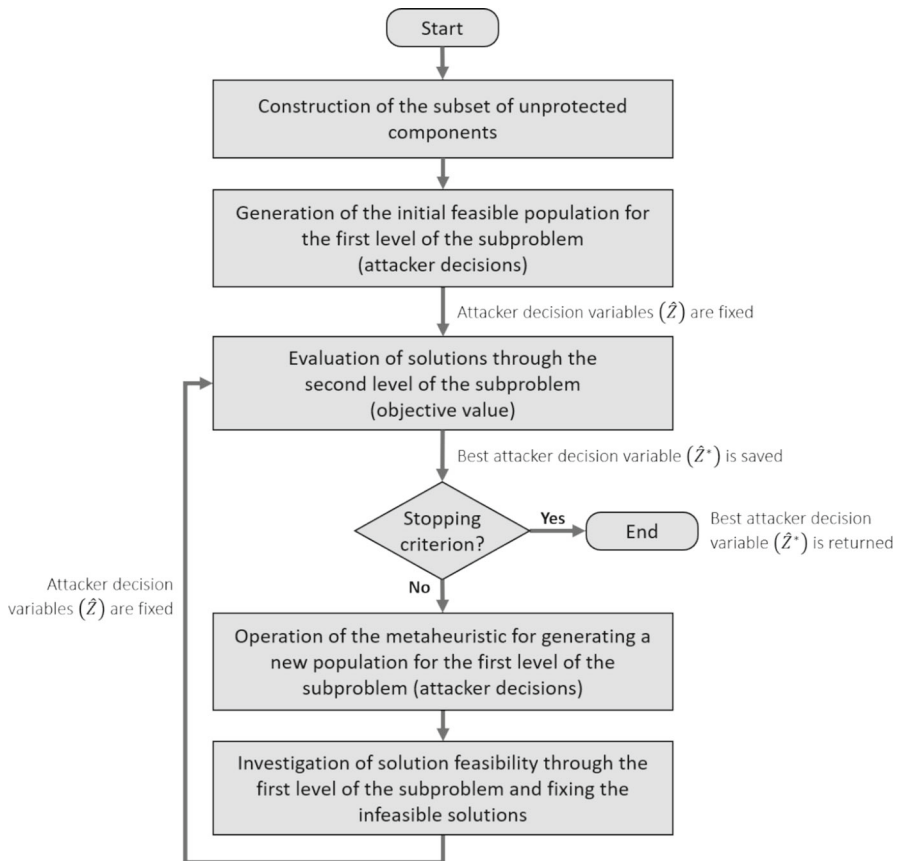


Fig. 2 A general framework of the proposed tailored metaheuristic algorithm for solving the subproblem

procedure explores the most critical components (which is not necessarily the optimal solution because of the heuristic nature of the algorithm) that the attacker should target to maximize the disruption subject to an interdiction budget. This framework reduces the expected required computation as it avoids checking components that are protected since the subset of unprotected components is the input of the metaheuristic algorithm. Note that if protector decision variables take binary values, construction of the subset of unprotected components is straightforward; otherwise, rules can be defined for constructing this subset (e.g., components that are protected less than 10%). After generating initial feasible attacker decisions, the quality of every solution is evaluated through the second level of the subproblem by calculating its corresponding objective value. Note that the quality of generated solutions is assessed from the perspective of the attacker, so a larger disruption is preferable. Then, the best attacker decision is saved for the procedure of the metaheuristic operation. A new set of solutions is generated through the metaheuristic operation, and generated solutions are evaluated again for the next iteration of the metaheuristic operation. Note that for every first level feasible solution (satisfying the first level constraints), there exists a feasible solution for the second level decision variables. In other words, for every attacker decision ($\hat{Z}$) satisfying the attacker budget constraint, the subsequent defender level has a feasible reaction. Therefore, every solution generated by the metaheuristic should be checked in terms of the feasibility criterion through the first level of the subproblem (generated solution should satisfy the attacker budget constraint). The metaheuristic algorithm continues until the stopping criterion is reached, and the best attacker decision is returned. This is an efficient solution but not necessarily the optimal solution because of the heuristic nature of the algorithm.

Based on the above discussion for addressing the master problem and subproblem, the implementation steps of the HBD algorithm are represented in Table 2. Note that ε is the solution gap set by the decision maker for the algorithm. LB , UB , obj_{MP} , and obj_{SP} refer to the lower bound, upper bound, and master problem (MP) and subproblem (SP) objective values, respectively. $\hat{Y}^*$ and $\hat{Z}^*$ denote the best (but not necessarily optimal) protection and interdiction decisions, respectively. Finally, $time_{start}$ and $time_{end}$ record the algorithm time.

At every iteration, the subproblem delivers an effective attack plan for a given protection scenario. The master problem is extended with the new attacker scenario accordingly. Since (i) The subproblem is solved with the tailored metaheuristic algorithm, and (ii) The subset of unprotected components is fed into the subproblem, a high-quality solution most likely will be returned in a shorter amount of time in comparison to the case of exploring the whole set of components. For more insight regarding the solution methods enumerating all possible combinations for the attacker plan, we refer the reader to Yao et al. (2007) and Ghorbani-Renani et al. (2020).

Note that for the case of solving both master problem and subproblem to optimality, as the algorithm proceeds, the upper and lower bound values of the formulation continue to converge until the final solution is obtained. However, since the subproblem is solved with a metaheuristic algorithm that might not offer the optimal attack strategy at each iteration, the final solution might not reflect the maximum capacity of the interdictor according to its resource availability. Hence, after the attacker and protector interactions, the subproblem might not provide a valid upper bound for the

Table 2 Implementation steps of HBD algorithm

HBD algorithm:	
Step 1	Initialization: $LB \leftarrow -\infty$, $UB \leftarrow +\infty$ and $\hat{Z}^c \leftarrow 0$ in $c \leftarrow 0$, $t \leftarrow 0$, $diff \leftarrow 0$
Step 2	$time_{start} = time$
Step 3	While $\frac{UB-LB}{LB} > \varepsilon$, $t < T$, and $diff < \mathcal{F}$ do:
Step 4	Solve MP, obtain its value obj_{MP} and protection decision $\hat{Y}$
Step 5	$\hat{Y}^* \leftarrow \hat{Y}$, $LB \leftarrow obj_{MP}$
Step 6	Solve SP, obtain its value obj_{SP} and interdiction decision $\hat{Z}$
Step 7	If $\min \{UB, obj_{SP}\} \geq LB$ then:
	$c \leftarrow c + 1$
	$\hat{Z}^* \leftarrow \hat{Z}$, $\hat{Z}^c \leftarrow \hat{Z}$, $obj_{SP}^* \leftarrow obj_{SP}$ $UB \leftarrow \min \{UB, obj_{SP}\}$
	$time_{end} = time$, $diff = time_{end} - time_{start}$
	Update MP by generating a new set of variables and constraints
	Else:
Step 8	Go to Step 6, $t \leftarrow t + 1$
Step 9	Exit (protection decision $\hat{Y}^*$, interdiction decision $\hat{Z}^*$, obj_{SP}^* are returned)

formulation. The subproblem objective value would then be less than the master problem objective value (e.g., lower bound values). Therefore, such circumstances lead to two main difficulties through the algorithm. The first and foremost issue is the early and incorrect termination of the algorithm based on the convergence stopping criterion (e.g., in such a case, since the upper bound value is less than the lower bound value, ε is negative which would be less than any solution gap, including 0, opted by the decision maker and the algorithm is terminated improperly). The second issue is that the master problem is updated at each iteration of the algorithm by receiving the new set of attacker decisions generated by the subproblem. However, as discussed previously, the metaheuristic algorithm might not offer the optimal attack strategy at each iteration, making the generated constraints (cuts) redundant for the master problem. This redundancy leads to exponential growth of the master problem by increasing the iteration index and raising the complexity of the problem unnecessarily without tightening the feasible region.

To solve this issue, we incorporate a new set of conditions to the algorithm disregarding the ineffective attacker plan suggested by the metaheuristic algorithm, otherwise the attacker strategy will be considered into the master problem for further procedures (referring to Steps 7 and 8). An “ineffective attacker plan” is identified when the metaheuristic algorithm generates an attacker strategy that results in a subproblem objective value that, when combined with the upper bound, is smaller than the lower bound. This condition is checked in Step 7 through the “if” statement. If this occurs, the solution is deemed ineffective, and the generated constraints (or cuts) become redundant for the master problem. As a result, the algorithm will resolve the metaheuristic again to produce a new attacker plan. If the new plan satisfies the necessary criteria, it will be

considered in the master problem. Otherwise, the previous solution will be discarded. With this effective modification, (i) The algorithm is not terminated improperly, (ii) The master problem is not expanded unnecessarily, and (iii) The algorithm finds an efficient solution (most likely the optimal solution) in a shorter amount of time. Therefore, by iteratively generating a new set of variables and constraints and solving the updated master problem, the upper and lower bound values of the formulation continue to converge until the final solution is obtained.

Although it would be a rare situation, we also consider the case where the metaheuristic algorithm returns several consecutive unimproved attacker solutions. In such a circumstance, the HBD algorithm is terminated if (i) The gap (ε) between the two bounds is sufficiently close, (ii) The number of times (T) that a consecutive lack of improvement in the attacker solutions is reached, as recorded in Step 8 of the algorithm, or (iii) The algorithm surpasses a prescribed time limit (e.g., $\mathcal{F}$).

Note that the proposed solution algorithm is applicable for either total or partial protection and interdiction decisions. For more clarification, in the binary version of the first defender and attacker decision variables, protected components will be totally immune to any attack. However, in the continuous case, in addition to partial decisions for the attacker, the invulnerability of the components is a function of its fortification.

4.2 Hybrid set covering-based (HSC) algorithm

As discussed in HBD algorithm in Sect. 4.1, the master problem is constructed based on the Benders framework. In some cases when the complexity of the master problem is high, as the algorithm iteration grows, the computational time for solving the model increases since a new set of decision variables and constraints are generated at each iteration. Therefore, we utilize the set covering approach to deal with the master problem instead of adding a large number of constraints to the model at each iteration. Hence, the HSC algorithm efficiently modifies the CD approach with the proposed tailored metaheuristic algorithm discussed in Sect. 4.1.

To convert the master problem to the single-level minimization, an inequality is generated and added to the problem at each iteration of the algorithm. This inequality forces the model to protect at least one interdicted component introduced by the attacker at every iteration. By adding this inequality at every iteration, the feasible region of the master problem is tightened until the model becomes infeasible and algorithm is terminated. In other words, due to the first defender budget constraint, at one iteration, the master problem cannot satisfy all inequalities added to the model, and the whole algorithm is terminated with the optimal protection decision. Then, to solve the subproblem, as with the HBD algorithm, we utilize the proposed tailored metaheuristic algorithm. However, doing so may result in an attacker strategy that might not be optimal. Therefore, the generated inequality might not be distinct from the previous inequalities, resulting in an unchanging feasible region of the master problem at some iterations. In such a case, the HSC algorithm is terminated if (i) The master problem becomes infeasible, (ii) A defined number of iterations (T) of consecutive unimproved solutions is reached, or (iii) The algorithm surpasses a prescribed time limit (e.g., $\mathcal{F}$).

Table 3 Implementation steps of HSC algorithm

HSC algorithm:	
Step 1	If protector budget is enough to protect all components, then : $\hat{Y}^* \leftarrow 1, \hat{Z}^* \leftarrow 0, UB \leftarrow 0$ and go to Step 10
Step 2	Initialization : $LB \leftarrow -\infty, UB \leftarrow +\infty, \hat{Y} \leftarrow 0, t \leftarrow 0, diff \leftarrow 0$
Step 3	$time_{start} = time$
Step 4	While MP is feasible, $t < T$, and $diff < \mathcal{F}$ do :
Step 5	Solve SP, obtain its value obj_{SP} and interdiction decision $\hat{Z}$
Step 6	If $obj_{SP} < UB$ then : $\hat{Y}^* \leftarrow \hat{Y}, \hat{Z}^* \leftarrow \hat{Z}, UB \leftarrow obj_{SP}$
Step 7	Add the following inequalities to MP: $\sum_{k \in K} \sum_{(i,j) \in A^k \hat{z}_{ij}^k = 1} y_{ij}^k \geq 1$
Step 8	Solve MP, obtain its value obj_{MP} and protection decision $\hat{Y}$
Step 9	If $obj_{MP} > LB$ then : $LB \leftarrow obj_{MP}, t \leftarrow 0$ $time_{end} = time, diff = time_{end} - time_{start}$ Else : $t \leftarrow t + 1$
Step 10	Exit (protection decision $\hat{Y}^*$ and interdiction decision $\hat{Z}^*$, and UB are returned)

Table 3 represents the implementation steps of the HSC algorithm. The definitions and abbreviations in Table 3 are similar to those in the HBD algorithm in Table 2. Step 9 of the algorithm records the number of consecutive unimproved solutions. Note that although the master problem in this method is easier to solve compared to the HBD algorithm and computational complexity is reduced in some cases, this approach is only applicable for the binary version of the first and second level decision variables.

5 Illustrative examples

To illustrate the efficient solvability of the proposed solution techniques, we study the existing DAD model, namely a tri-level protection-interdiction-restoration model developed by Ghorbani-Renani et al. (2020) and solve it for two case studies with different network sizes. Note that the devised model accounts for the restoration process of disrupted network components through the counteraction level as recourse decisions in the network resilience plan. Details of the formulation are presented in the appendix.

Case study I is the interdependent system of water, gas, and power networks in Shelby County, TN, USA, including 125 nodes and 164 links (Song & Ok 2010; Hernandez-Fajardo & Dueñas-Orsorio 2011; González et al. 2016). The topology of this system is depicted in Fig. 3, where the notation w, g, and p refer to water, gas, and power networks, respectively. Note that the dashed lines represent the interdependency relationship between water and power networks in this case study. To assess the performance of the proposed solution approaches for a larger problem, we generated an interdependent system of two networks including 252 nodes and 507 links, representing a double in the size of the Shelby County network. The simulated system was generated using the Networkx library in Python (Hagberg et al. 2008) according to the Barabási-Albert network model, thus not fully replicating the structure of the Shelby County network (NetworkX developers 2021). The main characteristic of this model is the preferential attachment, suggesting that the more connected a node is, the more likely it is to receive new links (Chassin & Posse 2005; Zadorozhnyi & Yudin 2014). Figure 4 demonstrates the topology of the simulated interdependent system of two networks that we assume to arbitrarily represent water and power networks for illustrative purposes. Similar to case study I, w and p refer to the water and power networks, respectively, and the dashed lines represent the interdependency relationship between the two networks in this system.

We implemented the optimization problems and solution algorithms in Python 3.6.5 with Gurobi 8.0.1. Results were conducted on a 64-bit operating system, Intel® Core™ i7-6700 CPU @ 3.40GHz 3.41GHz desktop computer. In the following subsections, the tri-level protection-interdiction-restoration formulation is briefly introduced, and the results are discussed in detail for both case studies.

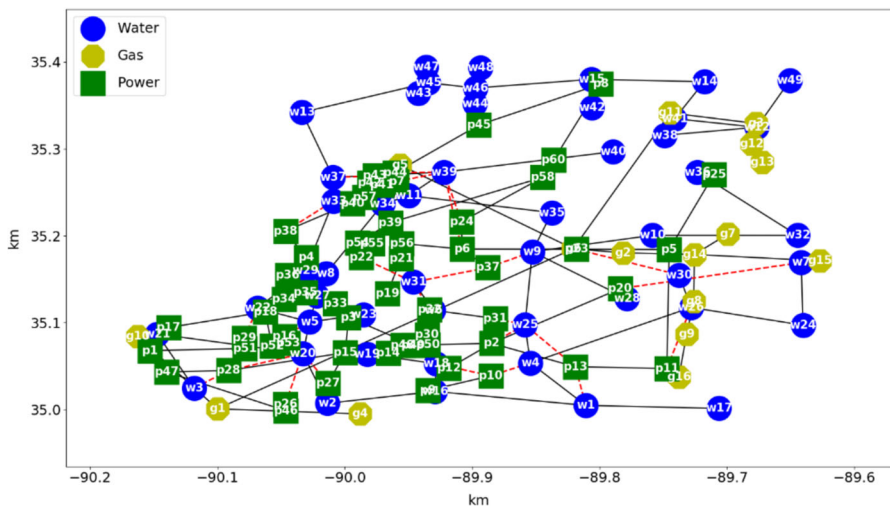


Fig. 3 The topology of the interdependent system of water, gas, and power networks in Shelby County, TN, USA: case study I

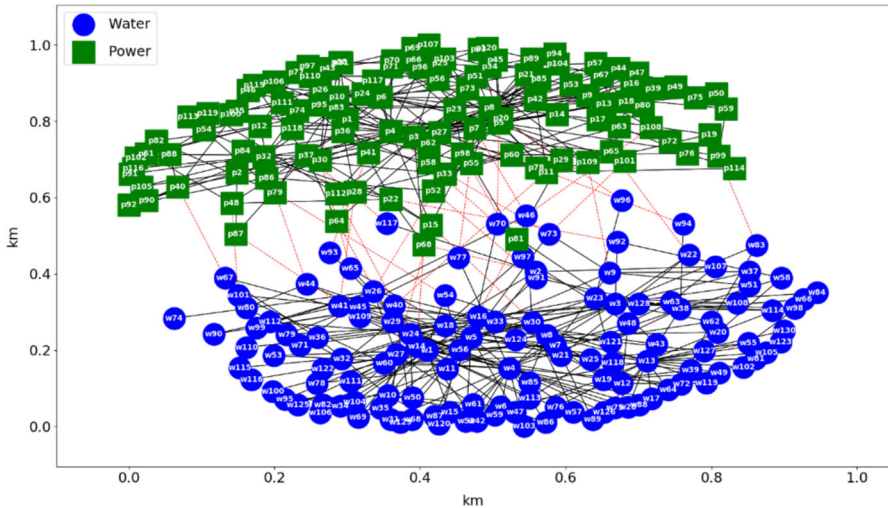


Fig. 4 The topology of the simulated interdependent system of water and gas networks: case study II

5.1 Computational results: case study I

In this section, we apply the proposed solution techniques to solve the tri-level protection-interdiction-restoration formulation developed by Ghorbani-Renani et al. (2020). To briefly review the model, three levels are involved in this formulation including (i) Protection, (ii) Interdiction, and (iii) Restoration. The protection level implements strategies for hardening network components. The interdiction level identifies the set of components, the disruption of which would result in the maximum damage through the system for a given attack budget. Finally, the restoration level is responsible for restoring the disrupted network to its baseline performance in a timely manner. Compared to the basic formulation presented in Sect. 3, the tri-level protection-interdiction-restoration formulation is more comprehensive as the recovery process of disrupted system is incorporated through the restoration level of the model. Note that the definition of the model sets, indices, parameters, and decision variables, as well as the formulation of the model, have been presented in detail in the appendix. Model parameters of the sample problems are selected according to Ghorbani-Renani et al. (2020).

To solve the tri-level protection-interdiction-restoration formulation with HBD and HSC algorithms discussed in Sects. 4.1 and 4.2, respectively, we apply the genetic algorithm in the subproblem. The parameters of the genetic algorithm are selected according to Arroyo and Fernandez (2009). The genetic algorithm is stopped when the number of generations is attained (Pelikan et al. 2002). Note that the convergence ratio of the upper and lower bounds in HBD algorithm (e.g., the stopping criterion of HBD algorithm) is set to $\varepsilon = 0.1$. Therefore, the HBD algorithm is terminated if either the gap between the two bounds is close enough (ε) or the number of times of returning consecutive unimproved solutions is reached (T). However, the HSC algorithm is

terminated if either the master problem becomes infeasible due to the protection budget constraint or the number of times of returning consecutive unimproved solutions is reached (T).

Two criteria are defined to assess the performance of the proposed solution approaches compared to the CD algorithm: (i) The optimality convergence ratio assesses the quality of the solution, and (ii) The computation time rate assesses computation speed, where ξ and Γ represent the objective value and the solution time (in minutes), respectively. The results of two case studies are discussed in detail subsequently.

Table 4 Performance evaluation of HBD algorithm with respect to the CD algorithm under different budget scenarios for case study I

Protector budget (BP)	Attacker budget (BI)	HBD algorithm		CD algorithm		$\left(\frac{\xi^{CD} - \xi^{HBD}}{\xi^{CD}}\right)$	$\left(\frac{\Gamma^{CD}}{\Gamma^{HBD}}\right)$
		ξ^{HBD}	$\Gamma^{HBD}(\text{min})$	ξ^{CD}	$\Gamma^{CD}(\text{min})$		
0	70	0.51	3.73	0.51	0.08	–	0.02
	140	0.87	3.81	0.87	0.42	–	0.11
	210	1.21	3.83	1.21	1.37	–	0.36
	280	1.78	4.19	1.78	3.63	–	0.87
	350	2.14	4.66	2.14	5.32	–	1.14
40	70	0.22	5.52	0.22	0.15	–	0.03
	140	0.53	5.65	0.53	1.19	–	0.21
	210	0.94	6.11	0.94	4.18	–	0.68
	280	1.21	6.40	1.21	12.74	–	1.99
	350	1.36	6.84	1.40	18.77	0.03	2.74
80	70	0.19	7.47	0.19	0.23	–	0.03
	140	0.45	11.35	0.45	1.56	–	0.14
	210	0.71	13.81	0.71	6.69	–	0.48
	280	0.90	14.76	0.90	29.40	–	1.99
	350	1.13	16.83	1.14	81.53	0.01	4.84
120	70	0.16	11.56	0.16	0.38	–	0.03
	140	0.37	20.11	0.37	3.13	–	0.16
	210	0.59	23.00	0.59	18.76	–	0.81
	280	0.80	49.05	0.80	100.87	–	2.06
	350	0.94	64.04	0.94	291.72	–	4.56
160	70	0.15	13.03	0.15	0.44	–	0.03
	140	0.32	21.04	0.32	4.73	–	0.23
	210	0.47	31.88	0.47	48.72	–	1.53
	280	0.61	65.40	0.61	269.82	–	4.13
	350	0.76	94.84	0.76	1021.61	–	10.77

Table 5 Performance evaluation of HSC algorithm with respect to the CD algorithm under different budget scenarios for case study I

Protector budget (BP)	Attacker budget (BI)	HSC algorithm		CD algorithm		$\left(\frac{\xi^{\text{CD}} - \xi^{\text{HSC}}}{\xi^{\text{CD}}}\right)$	$\left(\frac{\Gamma^{\text{CD}}}{\Gamma^{\text{HBD}}}\right)$
		ξ^{HSC}	$\Gamma^{\text{HSC}}(\text{min})$	ξ^{CD}	$\Gamma^{\text{CD}}(\text{min})$		
0	70	0.51	1.84	0.51	0.08	–	0.04
	140	0.87	1.95	0.87	0.42	–	0.22
	210	1.21	2.01	1.21	1.37	–	0.68
	280	1.78	2.07	1.78	3.63	–	1.75
	350	2.14	2.13	2.14	5.32	–	2.50
40	70	0.22	3.66	0.22	0.15	–	0.04
	140	0.53	3.79	0.53	1.19	–	0.31
	210	0.94	3.86	0.94	4.18	–	1.08
	280	1.21	10.04	1.21	12.74	–	1.27
	350	1.40	11.10	1.40	18.77	–	1.69
80	70	0.19	5.35	0.19	0.23	–	0.04
	140	0.45	9.14	0.45	1.56	–	0.17
	210	0.71	11.53	0.71	6.69	–	0.58
	280	0.87	19.82	0.90	29.40	–	1.48
	350	1.13	46.41	1.14	81.53	0.01	1.76
120	70	0.16	9.13	0.16	0.38	–	0.04
	140	0.37	13.04	0.37	3.13	–	0.24
	210	0.59	26.50	0.59	18.76	–	0.71
	280	0.80	73.55	0.80	100.87	–	1.37
	350	0.94	160.88	0.94	291.72	–	1.81
160	70	0.15	10.93	0.15	0.44	–	0.04
	140	0.32	18.17	0.32	4.73	–	0.26
	210	0.47	75.96	0.47	48.72	–	0.64
	280	0.61	161.45	0.61	269.82	–	1.67
	350	0.76	512.60	0.76	1021.61	–	1.99

The model is solved for different budget scenarios. Tables 4 and 5 show the efficiency of HBD and HSC algorithm, respectively, in comparison with the CD algorithm for different protector and attacker budget scenarios. According to the convergence criterion, both solution methods solve the model to optimality for almost all budget scenarios or the percentage of optimality gap is too small to be considered (e.g., 0.01, 0.03). Based on the results, it can be concluded that the proposed solution methods are successful to provide the high-quality solutions for such a complex model and are able to compete with exact solution methods in terms of the accuracy of the solution. Since the convergence ratio of upper bound and lower bound in HBD algorithm is set to 0.1 (e.g., $\varepsilon = 0.1$), the algorithm terminates when the stopping criterion is reached

therefore the returned solution might be less than the optimum solution returned by the CD algorithm.

Figure 5 depicts the computation time of each solution technique to solve the model under different budget scenarios for case study I. Note that to easily comprehend the comparison for smaller attacker budgets, we alter the vertical axes to a standard logarithmic scale, which is also applied to the remaining figures in this section. The trend indicates that the run times of all three algorithms are affected by the resources available to both protector and attacker. This is likely because, by increasing the budget, possible combinations (feasible solutions) for interdiction and protection grow, therefore the algorithms require more search effort to find the solution and terminate. According to Fig. 5, the run time of the CD algorithm is more affected by changes in budget relative to the HSC algorithm and then the HBD algorithm, whose run time demonstrates the least susceptibility to budget changes.

Further, with the CD algorithm, the attacker algorithm finds the best strategy based on the component evaluations in a systematic search (implicit enumeration), which guarantees optimality in a finite number of iterations (i.e., component evaluations). To provide further insight regarding this search algorithm, assume that $|N' \cup A'|$ represents the total number of candidate components subject to interdiction and the attacker can interdict up to n components (cardinality budget for the attacker). Therefore, the

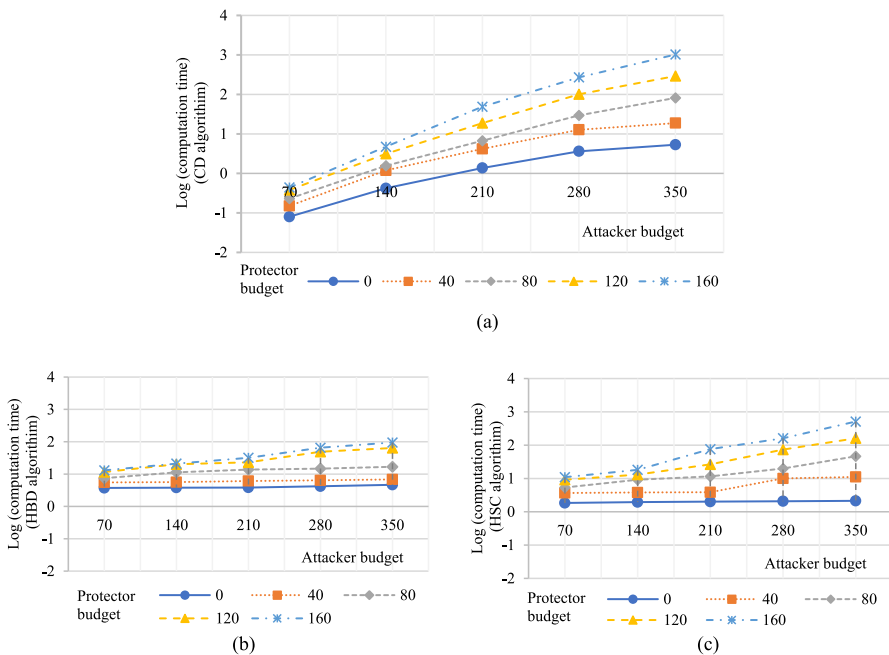


Fig. 5 Logarithm of computation time of **a** CD, **b** HBD, and **c** HSC algorithms under different budget scenarios for case study I

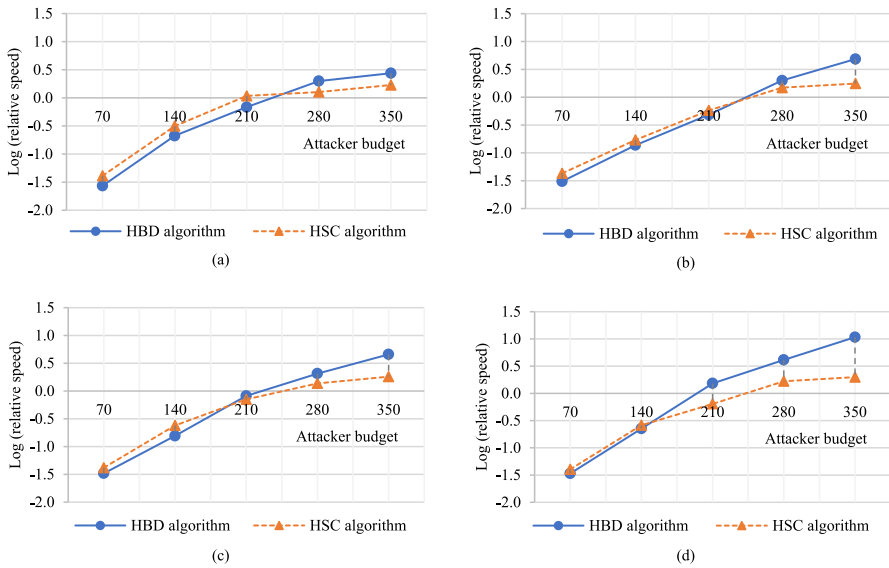


Fig. 6 Logarithm of relative computation speed of HBD and HSC algorithms with respect to CD algorithm for **a** $BP = 40$, **b** $BP = 80$, **c** $BP = 120$, and **d** $BP = 160$ for case study I

total number of node evaluations in this search is $\left(\frac{|N' \cup A'|}{Optn}\right)$, which grows exponentially with a larger number of candidate components and higher available budget. This is where the advantages of the metaheuristic algorithm, namely the proposed hybrid solution approaches, stand out since all possible combinations are not evaluated.

Figure 6 illustrates the computation speed of each algorithm with respect to the CD algorithm under different budget scenarios. Mentioned previously, we consider the standard logarithm of relative speed to enable comparing the result for smaller attacker budgets. The trend reveals that, in general, both HBD and HSC algorithms significantly outperform the CD algorithm with respect to computation time, particularly for higher budget scenarios. For example, for $BP = 80$ and $BI = 350$, the HBD algorithm method solves the model to optimality in about 16 min, and HSC method finds the solution in around 46 min with an optimality gap ratio of 0.01. This compares with 81 min to find the optimal solution with the CD algorithm, which is approximately 5 times slower than HBD and 2 times slower than HSC. For the scenario of $BP = 160$ and $BI = 350$, the computation speed of HBD and HSC algorithms are respectively around 11 and 2 times faster than the CD algorithm. Although the CD algorithm solves the model to optimality, the computational performance of the algorithm decreases exponentially by increasing the size of the problem or the amount of available budget, which is the main shortcoming of the CD solution method. Note that for the lower budget scenarios, the CD algorithm finds the solutions slightly faster than the other two solution methods, but the differences in computation times of all three algorithms are negligible. In addition, the computational complexity of DAD models emerges when we are dealing with a large-sized problem with higher budget availabilities therefore

an algorithm is known to be efficient if it can find the solution for such circumstances with an acceptable accuracy in a reasonable time.

Furthermore, the results indicate that HBD algorithm surpasses HSC in terms of computation speed in case study I. Although the master problem in the second solution algorithm is easier to solve in comparison with the Benders approach in the first algorithm, it requires more iterations with the addition of new cuts to the model until it becomes infeasible. At each iteration of the algorithm, the best protector solution is recorded by comparing the objective value with the previous iteration (referring to steps 4 and 5 in Table 3). Therefore, it may require more iterations relative to Benders approach. With Benders approach, the master problem at each iteration finds the best protection plan by considering all attack strategies proposed to that point, leading to the growth of the master problem by increasing the iteration index. Therefore, there is a tradeoff between the complexity of the master problem in HBD algorithm and the number of iterations of HSC algorithm.

Table 6 illustrates the protector (SP), and attacker (SI) strategies resulted from 3 solution algorithms (CD, HBD, and HSC) under protector budgets of 120 and 160 and attacker budgets ranging from 70 to 350. The notation (k, i) used to describe the network components refers to node $i \in N'^k$ in network $k \in K$, where water, gas, and power networks are denoted by 1, 2, and 3, respectively. Based on the results, network components (1, 5), (1, 26), (2, 14), (3, 2), and (3, 6) are protected in most budget scenarios representing critical components in the system. Such a sensitivity analysis provides beneficial insights for generating set of inequalities making the model to protect given network components considering the budget availability. In addition, such inequalities enhance the efficiency of the algorithms for higher budget scenarios.

Table 6 Protector and attacker decisions under different budget scenarios

$BP=120$	$BI=70$	$BI=140$	$BI=210$	$BI=280$	$BI=350$
SI	(3,6)	(3,6), (3,15)	(1,5), (2,6), (2,14)	(1,5), (2,6), (2,14), (1,9)	(1,5), (2,6), (2,14), (1,9), (3,15)
SP	(1,5), (1,26), (2,14), (3,2)	(1,5), (1,26), (2,14), (3,2)	(1,26), (3,2), (1,16), (3,15)	(1,26), (3,2), (1,16), (3,6)	(1,26), (3,2), (1,16), (3,6)
$BP=160$	$BI=70$	$BI=140$	$BI=210$	$BI=280$	$BI=350$
SI	(2,6)	(1,9), (1,16)	(1,9), (1,16), (2,6)	(1,9), (1,16), (3,7), (3,15)	(1,9), (1,16), (3,7), (3,15), (2,6)
SP	(1,5), (1,26), (2,14), (3,2), (3,6)	(1,5), (1,26), (2,14), (3,2), (3,6)	(1,5), (1,26), (2,14), (3,2), (3,6)	(1,5), (1,26), (2,14), (3,2), (3,6)	(1,5), (1,26), (2,14), (3,2), (3,6)

Table 7 Performance evaluation of HBD algorithm with respect to the CD algorithm under different budget scenarios for case study II

Protector budget (BP)	Attacker budget (BI)	HBD algorithm		CD algorithm		$\left(\frac{\xi^{CD} - \xi^{HBD}}{\xi^{CD}} \right)$	$\left(\frac{\Gamma^{CD}}{\Gamma^{HBD}} \right)$
		ξ^{HBD}	$\Gamma^{HBD}(\min)$	ξ^{CD}	$\Gamma^{CD}(\min)$		
0	70	0.26	10.46	0.26	0.25	–	0.02
	140	0.58	10.59	0.58	1.22	–	0.12
	210	0.94	11.09	0.94	4.05	–	0.37
	280	1.51	25.86	1.51	9.23	–	0.36
	350	2.14	29.21	2.14	15.7	–	0.54
40	70	0.21	15.64	0.21	0.47	–	0.03
	140	0.43	26.76	0.43	3.50	–	0.13
	210	0.78	33.80	0.78	15.88	–	0.47
	280	1.18	57.21	1.18	52.82	–	0.92
	350	1.59	178.96	1.69	90.4	0.06	0.51
80	70	0.18	21.53	0.18	0.70	–	0.03
	140	0.39	44.14	0.39	4.78	–	0.11
	210	0.73	47.78	0.73	19.57	–	0.41
	280	1.04	125.46	1.04	81.72	–	0.65
	350	1.30	115.47	1.30	247.45	–	2.14
120	70	0.15	26.92	0.15	0.92	–	0.03
	140	0.35	61.46	0.35	10.25	–	0.17
	210	0.52	69.45	0.52	59.03	–	0.85
	280	0.67	80.60	0.67	239.20	–	2.97
	350	0.86	98.49	0.86	718.75	–	7.30
160	70	0.12	31.92	0.12	1.13	–	0.04
	140	0.26	83.55	0.26	20.00	–	0.24
	210	0.44	131.25	0.44	124.63	–	0.95
	280	0.55	178.11	0.55	618.45	–	3.47
	350	0.68	381.59	0.68	2758.20	–	7.23

5.2 Computational results: case study II

Tables 7 and 8 represent the performance of HBD and HSC algorithms, respectively, compared to the CD algorithm for case study II. Although the size of the network is doubled, both hybrid algorithms perform efficiently in providing high quality results as the optimum solution is returned for almost all scenarios or otherwise the optimality gap is negligible (e.g., 0.05, 0.06).

As a result of increasing the size of the case study, computation times of all three solution algorithms naturally rise for all budget scenarios. Figure 7 illustrates the behavior of computational time by increasing the budget availability for each solution algorithm for case study II. Like case study I, we use the standard logarithmic scale

Table 8 Performance evaluation of HSC algorithm with respect to the CD algorithm under different budget scenarios for case study II

Protector budget (BP)	Attacker budget (BI)	HSC algorithm		CD algorithm		$\left(\frac{\xi^{\text{CD}} - \xi^{\text{HSC}}}{\xi^{\text{CD}}}\right)$	$\left(\frac{\Gamma^{\text{CD}}}{\Gamma^{\text{HBD}}}\right)$
		ξ^{HSC}	$\Gamma^{\text{HBD}}(\text{min})$	ξ^{CD}	$\Gamma^{\text{CD}}(\text{min})$		
0	70	0.26	5.34	0.26	0.25	–	0.05
	140	0.58	5.48	0.58	1.22	–	0.22
	210	0.94	5.86	0.94	4.05	–	0.69
	280	1.51	6.39	1.51	9.23	–	1.44
	350	2.14	7.64	2.14	15.7	–	2.06
40	70	0.21	11.10	0.21	0.47	–	0.04
	140	0.43	16.83	0.43	3.50	–	0.21
	210	0.78	23.80	0.78	15.88	–	0.67
	280	1.18	31.81	1.18	52.82	–	1.66
	350	1.60	44.20	1.69	90.4	0.05	2.05
80	70	0.18	16.23	0.18	0.70	–	0.04
	140	0.39	27.64	0.39	4.78	–	0.17
	210	0.73	28.37	0.73	19.57	–	0.69
	280	1.04	68.87	1.04	81.72	–	1.19
	350	1.30	134.72	1.30	247.45	–	1.84
120	70	0.15	22.13	0.15	0.92	–	0.04
	140	0.35	43.96	0.35	10.25	–	0.23
	210	0.52	82.91	0.52	59.03	–	0.71
	280	0.67	165.57	0.67	239.20	–	1.45
	350	0.86	509.68	0.86	718.75	–	1.41
160	70	0.12	26.11	0.12	1.13	–	0.04
	140	0.26	69.91	0.26	20.00	–	0.29
	210	0.44	196.28	0.44	124.63	–	0.64
	280	0.55	484.75	0.55	618.45	–	1.28
	350	0.68	1396.73	0.68	2758.20	–	1.98

for the vertical axes of the figures to clearly see the comparison for smaller attacker budgets. In particular, this trend signifies how the budget change affects the growth of computational burden for each solution algorithm. Like case study I, we see the substantial sensitivity of the CD algorithm to changes in budget, highlighting the deficiency of this method for higher budget scenarios, as shown in Fig. 7a. However, like case study I, the computational complexity of the HBD algorithm is not greatly influenced by the change of budget. This is also true, though less so, for the HSC algorithm.

Figure 8 illustrates the logarithm of computation time of each algorithm relative to the CD algorithm. As shown in the first case as well, the computational times are generally sensitive to the budget availability. In addition, the upward trends for all

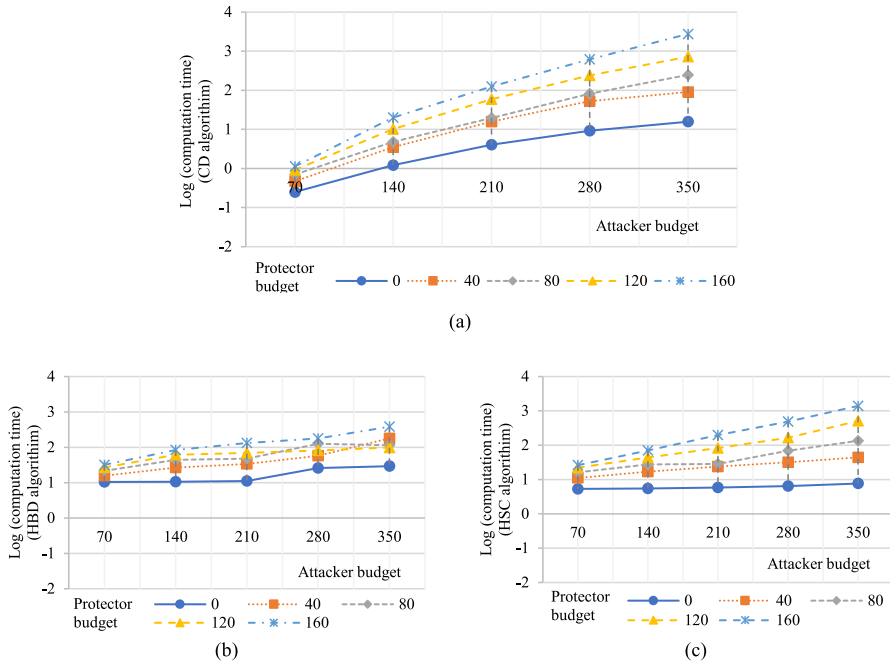


Fig. 7 Logarithm of computation time of **a** CD, **b** HBD, and **c** HSC algorithms under different budget scenarios for case study II

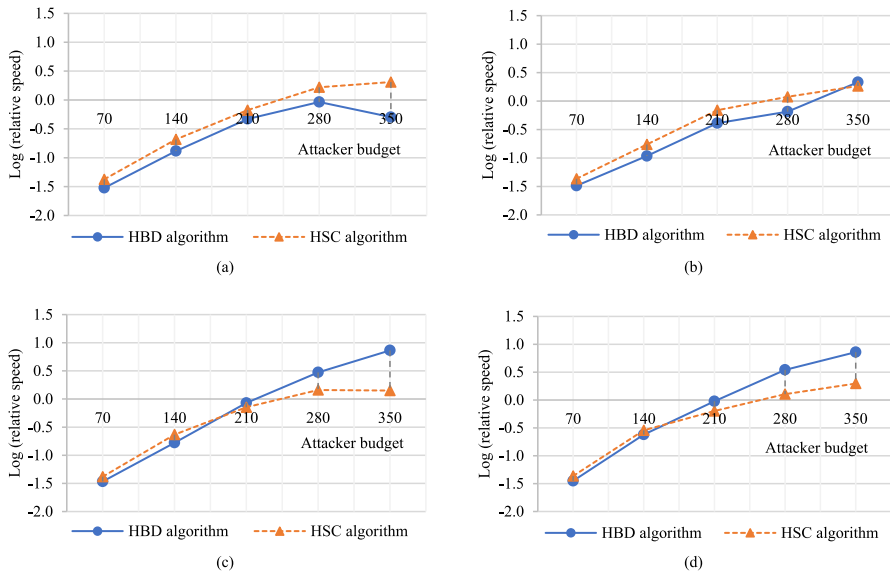


Fig. 8 Logarithm of relative computation speed of HBD and HSC algorithms with respect to CD algorithm for **a** $BP = 40$, **b** $BP = 80$, **c** $BP = 120$, and **d** $BP = 160$ for case study II

budget scenarios indicate the advantages of both HBD and HSC algorithms relative to the CD algorithm in terms of computational burden. Particularly for a given protection budget, as the attacker resource continues to grow, the supremacy of the proposed solution approaches in comparison to the CD algorithm enhances remarkably. Like case study I, the HBD algorithm is superior to HSC in terms of computation speed in case study II for higher budget scenarios, as shown in Fig. 8c and d. However, for the lower budget scenarios, Fig. 8a and b suggest that the HSC algorithm outperforms the HBD algorithm. This is because of the significant growth in the complexity of Benders decomposition algorithm applied in HBD algorithm as a result of increasing the size of the network. As we discussed in Sect. 4.1, at each iteration of HBD algorithm, the master problem is updated by receiving the new set of attacker decisions generated by the subproblem leading to exponential growth of the master problem by increasing the iteration index. However, in the HSC algorithm, the set covering algorithm is applied to solve the master problem, in which at each iteration only one inequality (cut) will be added to the master problem, enforcing that at least one interdicted component from the recent attacker strategy should be defended. On the other hand, the HSC algorithm requires more iterations to be terminated, which leads to the HBD algorithm outperforming it for higher budget scenarios, shown in Fig. 8c and d.

In our comparison across different budget scenarios, we find that the computational efficiency of both the HBD and HSC algorithms improves as the available budget grows. In lower budget scenarios, the performance of both algorithms is similar, with the HSC algorithm showing slightly better performance in terms of computational speed. This is because the HSC algorithm only adds one inequality (cut) per iteration, enabling it to terminate faster. In contrast, the Benders decomposition used in the HBD algorithm results in exponential growth of the master problem, increasing its complexity as the number of iterations increases, which makes the HBD algorithm less efficient in lower budget scenarios.

However, as the available budget increases, the number of iterations required for termination in the HSC algorithm also increases, leading to a higher computational burden. On the other hand, the HBD algorithm becomes more efficient in higher budget scenarios as it requires fewer iterations to terminate and reach a solution. This makes the HBD algorithm more computationally efficient in scenarios with larger budgets compared to the HSC algorithm, which struggles with the growing number of iterations.

Both the HBD and HSC algorithms show clear advantages over the set covering (CD) approach, particularly as the available budget increases. The CD algorithm experiences a notable increase in computational burden as the available budget grows, making it less efficient compared to the proposed algorithms. The logarithmic computation times of the HBD and HSC algorithms demonstrate that, as the available budget increases, both algorithms outperform the CD approach, with the HBD algorithm showing superior efficiency in higher budget scenarios.

In summary, the choice between the HBD and HSC algorithms depends on the available budget. For lower budget scenarios, the performance of both algorithms is similar, with a slight advantage for the HSC algorithm in terms of computational speed. However, in higher budget scenarios, the HBD algorithm becomes more computationally efficient and is the preferred choice.

Note that the accuracy of the proposed solution approaches is affected by the parameters designated for the metaheuristic algorithm, which clearly alter the computational speed of the solution techniques. For example, an additional stopping criterion can be defined for the genetic algorithm in which the difference of the solutions returned by two consecutive generations is zero. Doing so could significantly reduce the run time of the genetic algorithm since it may reach a good solution much earlier than the number of generations allowed. Therefore, this stopping criterion avoids the repetition of the same solution until the number of generations is obtained. However, it might affect the efficiency of the solution in some cases. In addition, due to the heuristic nature of the proposed solution algorithms, the reported solutions might not serve value of the optimal solution. Therefore, sensitivity analysis provides beneficial insights with respect to critical components in the system that involves solving the model with an exact solution algorithm (e.g., CD algorithm and C&CG) for several lower budget scenarios (refer to Table 6). Hence, the output can serve as valid inequalities to the model and thwart catastrophic damage of the attacks.

6 Concluding remarks

Interdiction models are generally a form of multi-level problems including a series of nested optimization models controlled by a set of constraints and decision variables in which the value of decision variables at each level affect the objective function of other levels (Rahdar 2016). This kind of model has a wide variety of real-world applications such as defense and military applications, traffic management, and infrastructure networks. However, their NP-hard nature makes them challenging to solve, particularly for moderate- to large-scale systems. The complexity of these models has made them a focus of extensive research, with the development of effective solution techniques remaining a crucial challenge.

As such, this study addressed this issue by proposing two hybrid solution techniques to overcome the computational complexity of network interdiction models, particularly DAD formulations. This study has provided a comprehensive framework for solving a wide range of network-based tri-level models. The proposed solution approaches are decomposition-based, so tri-level models are broken down into two nested formulations including master problem and subproblem. Two methods are combined to deal with the nested problems (master problem and subproblem) iteratively, and subsequently the hybrid solution algorithm is constructed to find the solution for the whole tri-level model. The first solution approach is the HBD algorithm, which is a novel extension of Benders decomposition method with a tailored metaheuristic algorithm, dealing with the network-based tri-level formulations with different model characteristics. The main advantage of this solution technique is its applicability to a broad variety of interdiction models with integer or continuous decision variables. The second approach uniquely modifies the CD method with the proposed tailored metaheuristic algorithm, referred to here as the HSC algorithm.

The proposed solution techniques were demonstrated with the protection-interdiction-restoration model developed by Ghorbani-Renani et al. (2020) as applied to two case studies with different network sizes: (i) The interdependent system of

water, gas, and power networks in Shelby County, TN, USA including 125 nodes and 164 links, and (ii) The simulated interdependent system of two networks, including 252 nodes and 507 links. Both the HBD and HSC algorithms perform well across various budget scenarios in the tri-level DAD model, often providing optimal or near-optimal solutions. In lower budget cases, their performance is similar, with the HSC algorithm being slightly faster due to its simpler iteration process. However, the HBD algorithm, using Benders decomposition, faces increased complexity and slower performance in these cases. As the budget increases, the HSC algorithm requires more iterations, becoming less efficient, while the HBD algorithm becomes more effective, requiring fewer iterations to reach a solution. Both algorithms outperform the CD algorithm, with the HBD algorithm showing superior efficiency in higher budget scenarios. The choice between the two depends on the budget, with the HSC algorithm being preferable for lower budgets and the HBD algorithm for higher budgets.

The strength of these proposed hybrid techniques lies in their ability to efficiently solve complex, large-scale tri-level interdiction problems, making them highly relevant for real-world applications. These techniques are flexible and can be adapted to a wide range of practical scenarios, including network security, infrastructure protection, and defense planning. The ability to handle both integer and continuous decision variables further enhances their applicability, allowing them to address a variety of real-world challenges. A promising direction for future work is the adaptation of these hybrid approaches to tackle NP-hard non-flow-based interdiction problems, such as knapsack and clustering problems. These types of problems, often encountered in real-world applications, can benefit from the integration of additional metaheuristics or adjustments to the algorithmic structure of the HBD and HSC frameworks. Exploring these adaptations would significantly enhance the practical relevance of these methods, extending their applicability to a broader range of complex interdiction models. Moreover, further developments could incorporate real-world complexities such as uncertainty, cascading failures, and imperfect information sharing (e.g., defender secrecy). Incorporating network-specific physical characteristics—such as Kirchhoff's laws for electrical networks or Pascal's principle for water networks—would enhance the relevance of these models in specific infrastructure domains.

In conclusion, this study makes a valuable contribution to the field of multi-level interdiction models by presenting two hybrid solution techniques that improve computational efficiency and broaden the applicability of these models to real-world scenarios. The proposed methods offer a significant promise for solving large-scale, complex interdiction problems. Future work should focus on adapting these methods for other NP-hard problems, incorporating uncertainty, and exploring the strategic role of information sharing to further enhance the practical impact of these solutions. Additionally, future work would benefit from exploring the application of reinforcement learning for controlling risk and resilience in interdependent infrastructure systems, providing a promising avenue for addressing optimization challenges in such systems.

Appendix: Tri-level formulation: protection-interdiction-restoration model

An undirected network is denoted by $G = (N, A)$, where N is the set of nodes, and A is the set of links. Assume a set K of networks, each with a set of nodes N^k such that $\bigcup_{k \in K} N^k = N$ and set of links A^k such that $\bigcup_{k \in K} A^k = A$. Nodes can consist of supply nodes ($N_+^k \subseteq N^k$), demand nodes ($N_-^k \subseteq N^k$) and transshipment nodes ($N_{=}^k \subseteq N^k \setminus \{N_+^k, N_-^k\}$) such that $N_+^k \cap N_-^k = \emptyset$. Sets $N'^k \subseteq N^k$ and $A'^k \subseteq A^k$ are candidate nodes and links, respectively, in network $k \in K$, that can be interdicted or protected in the system of interdependent networks. Ψ represents interdependency among networks such that $((i, k), (\bar{i}, \bar{k})) \in \Psi$ denotes node $i \in N^k$ in network $k \in K$ physically depends on node $\bar{i} \in N^{\bar{k}}$ in network $\bar{k} \in K$ where $N^k \cap N^{\bar{k}} = \emptyset$, $A^k \cap A^{\bar{k}} = \emptyset$ and $\forall k, \bar{k} \in K : k \neq \bar{k}$. Set R^k represents the available work crews in network $k \in K$. Index $t \in T$ provides the set of available time periods. Tables 9 and 10 outline the model parameters and decision variables, respectively.

See Tables 9 and 10.

Table 9 Model parameters

w_{it}^k	Importance weight assigned to node $i \in N_-^k$ in network $k \in K$ at time $t \in T$
CP_{ij}^k	Cost of protecting link $(i, j) \in A'^k$ in network $k \in K$
CP_i^k	Cost of protecting node $i \in N'^k$ in network $k \in K$
CI_{ij}^k	Cost of attacking link $(i, j) \in A'^k$ in network $k \in K$
CI_i^k	Cost of attacking node $i \in N'^k$ in network $k \in K$
BP	Total available budget for the protector
BI	Total available budget for the interdictor
s_i^k	Amount of supply in node $i \in N_+^k$ in network $k \in K$
d_i^k	Amount of demand in node $i \in N_-^k$ in network $k \in K$
u_{ij}^k	Capacity of link $(i, j) \in A^k$ in network $k \in K$
λ_{ij}^k	Restoration rate of the link $(i, j) \in A'^k$ in network $k \in K$
λ_i^k	Restoration rate of the node $i \in N'^k$ in network $k \in K$
ε	An arbitrarily small positive number, $0 < \varepsilon < 1$
M	An arbitrarily large positive number

Table 10 Model decision variables

m_{it}^k	Amount of demand met at node $i \in N_-^k$ in network $k \in K$ at time $t \in T$, continuous
x_{ijt}^k	Flow on link $(i, j) \in A^k$ in network $k \in K$ in time $t \in T$, continuous
y_{ij}^k	Equal to 1 if link $(i, j) \in A'^k$ in network $k \in K$ is protected, binary
y_i^k	Equal to 1 if node $i \in N'^k$ in network $k \in K$ is protected, binary
z_{ij}^k	Equal to 1 if link $(i, j) \in A'^k$ in network $k \in K$ is interdicted, binary
z_i^k	Equal to 1 if node $i \in N'^k$ in network $k \in K$ is interdicted, binary
f_{ij}^k	Equal to 1 if link $(i, j) \in A'^k$ in network $k \in K$ is disrupted, binary
f_i^k	Equal to 1 if node $i \in N'^k$ in network $k \in K$ is disrupted, binary
α_{ij}^k	Equal to 1 if link $(i, j) \in A'^k$ in network $k \in K$ is operational, binary
α_i^k	Equal to 1 if node $i \in N'^k$ in network $k \in K$ is operational, binary
α'_{ijr}	Equal to 1 if link $(i, j) \in A'^k$ in network $k \in K$ is restored by work crew $r \in R^k$ in time $t \in T$, binary
α'_{itr}	Equal to 1 if node $i \in N'^k$ in network $k \in K$ is restored by work crew $r \in R^k$ in time $t \in T$, binary
β_{ijt}^k	Equal to 1 if link $(i, j) \in A'^k$ in network $k \in K$ is reactivated at time $t \in T$, binary
β_{it}^k	Equal to 1 if node $i \in N'^k$ in network $k \in K$ is reactivated at time $t \in T$, binary

$$\zeta(t) = \frac{\sum_{k \in K} \sum_{i \in N_-^k} w_{it}^k (d_i^k - m_{it}^k)}{\sum_{k \in K} \sum_{i \in N_-^k} w_{it}^k d_i^k} \quad \forall t \in T \quad (19)$$

$$\xi = \min_y \max_z \min_{m, x, f, \alpha, \alpha', \beta} \sum_{t \in T} \zeta(t) \quad (20)$$

Subject to :

$$\sum_{k \in K} \sum_{(i, j) \in A^k} C P_{ij}^k y_{ij}^k + \sum_{k \in K} \sum_{i \in N'^k} C P_i^k y_i^k \leq BP \quad (21)$$

$$\sum_{k \in K} \sum_{(i, j) \in A'^k} C I_{ij}^k z_{ij}^k + \sum_{k \in K} \sum_{i \in N'^k} C I_i^k z_i^k \leq BI \quad (22)$$

$$y_{ij}^k + (1 - z_{ij}^k)(1 - y_{ij}^k) = 1 - f_{ij}^k \quad \forall (i, j) \in A'^k, \forall k \in K \quad (23)$$

$$y_i^k + (1 - z_i^k)(1 - y_i^k) = 1 - f_i^k \quad \forall i \in N'^k, \forall k \in K \quad (24)$$

$$\alpha_{ij}^k \leq 1 - f_i^k \quad \forall (i, j) \in A'^k, \forall k \in K \quad (25)$$

$$\alpha_{ij}^k + f_{ij}^k \geq \varepsilon \quad \forall (i, j) \in A'^k, \forall k \in K \quad (26)$$

$$\alpha_i^k \leq 1 - f_i^k \quad \forall i \in N'^k, \forall k \in K \quad (27)$$

$$\alpha_i^k + f_i^k \geq \varepsilon \quad \forall i \in N'^k, \forall k \in K \quad (28)$$

$$\alpha_{ijt}^k \leq 1 - \beta_{ijt}^k \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T \quad (29)$$

$$\alpha_i^k \leq 1 - \beta_{it}^k \quad \forall i \in N'^k, \forall k \in K, \forall t \in T \quad (30)$$

$$\beta_{ij1}^k = 0 \quad \forall (i, j) \in A'^k, \forall k \in K \quad (31)$$

$$\beta_{i1}^k = 0 \quad \forall i \in N'^k, \forall k \in K \quad (32)$$

$$\sum_{(i,j) \in A^k} x_{ijt}^k - \sum_{(j,i) \in A^k} x_{jit}^k \leq s_i^k \quad \forall i \in N_+^k, \forall k \in K, \forall t \in T \quad (33)$$

$$\sum_{(i,j) \in A^k} x_{ijt}^k - \sum_{(j,i) \in A^k} x_{jit}^k = 0 \quad \forall i \in N_-^k, \forall k \in K, \forall t \in T \quad (34)$$

$$\sum_{(i,j) \in A^k} x_{ijt}^k - \sum_{(j,i) \in A^k} x_{jit}^k = -m_{it}^k \quad \forall i \in N_-^k, \forall k \in K, \forall t \in T \quad (35)$$

$$m_{it}^k \leq d_i^k \quad \forall i \in N_-^k, \forall k \in K, \forall t \in T \quad (36)$$

$$x_{ijt}^k \leq u_{ij}^k \quad \forall (i, j) \in A^k, \forall k \in K, \forall t \in T \quad (37)$$

$$x_{ijt}^k \leq u_{ij}^k (\alpha_{ij}^k + \beta_{ijt}^k) \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T \quad (38)$$

$$x_{ijt}^k \leq u_{ij}^k (\alpha_i^k + \beta_{it}^k) \quad \forall (i, j) \in A^k, \forall k \in K, \forall i \in N'^k, \forall t \in T \quad (39)$$

$$x_{ijt}^k \leq u_{ij}^k (\alpha_j^k + \beta_{jt}^k) \quad \forall (i, j) \in A^k, \forall k \in K, \forall j \in N'^k, \forall t \in T \quad (40)$$

$$\sum_{s=1}^t \alpha'_{ijs} \leq M \left(1 - (\alpha'_{ij,t+1} - \alpha'_{ijt}) \right) \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T, \forall r \in R^k \quad (41)$$

$$\sum_{r \in R^k} \sum_{t \in T} \alpha'_{ijt} \geq \frac{f_{ij}^k}{\lambda_{ij}^k} - M \alpha_{ij}^k \quad \forall (i, j) \in A'^k, \forall k \in K \quad (42)$$

$$\sum_{r \in R^k} \sum_{t \in T} \alpha'_{ijt}{}^{kr} < \left(\frac{f_{ij}^k}{\lambda_{ij}^k} + 1 \right) + M\alpha_{ij}^k \quad \forall (i, j) \in A'^k, \forall k \in K \quad (43)$$

$$\sum_{s=1}^t \alpha'_{is}{}^{kr} \leq M \left(1 - (\alpha'_{i,t+1}{}^{kr} - \alpha'_{it}{}^{kr}) \right) \quad \forall i \in N'^k, \forall k \in K, \forall t \in T, \forall r \in R^k \quad (44)$$

$$\sum_{r \in R^k} \sum_{t \in T} \alpha'_{it}{}^{kr} \geq \frac{f_i^k}{\lambda_i^k} - M\alpha_i^k \quad \forall i \in N'^k, \forall k \in K \quad (45)$$

$$\sum_{r \in R^k} \sum_{t \in T} \alpha'_{it}{}^{kr} < \left(\frac{f_i^k}{\lambda_i^k} + 1 \right) + M\alpha_i^k \quad \forall i \in N'^k, \forall k \in K \quad (46)$$

$$\frac{\sum_{r \in R^k} \sum_{s=1}^{t-1} \alpha'_{ijs}{}^{kr}}{f_{ij}^k / \lambda_{ij}^k} \geq \beta_{ijt}^k \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T | t \neq 1 \quad (47)$$

$$\frac{\sum_{r \in R^k} \sum_{s=1}^{t-1} \alpha'_{is}{}^{kr}}{f_i^k / \lambda_i^k} \geq \beta_{it}^k \quad \forall i \in N'^k, \forall k \in K, \forall t \in T | t \neq 1 \quad (48)$$

$$\sum_{\substack{s \in R^k \\ s \neq r}} \sum_{t \in T} \alpha'_{ijt}{}^{ks} \leq M \left(1 - \alpha'_{ijt}{}^{kr} \right) \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T, \forall r \in R^k \quad (49)$$

$$\sum_{\substack{s \in R^k \\ s \neq r}} \sum_{t \in T} \alpha'_{it}{}^{ks} \leq M \left(1 - \alpha'_{it}{}^{kr} \right) \quad \forall i \in N'^k, \forall k \in K, \forall t \in T, \forall r \in R^k \quad (50)$$

$$\sum_{r \in R^k} \alpha'_{ijt}{}^{kr} \leq 1 \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T \quad (51)$$

$$\sum_{r \in R^k} \alpha'_{it}{}^{kr} \leq 1 \quad \forall i \in N'^k, \forall k \in K, \forall t \in T \quad (52)$$

$$\sum_{(i,j) \in A'^k} \alpha'_{ijt}{}^{kr} + \sum_{i \in N'^k} \alpha'_{it}{}^{kr} \leq 1 \quad \forall k \in K, \forall t \in T, \forall r \in R^k \quad (53)$$

$$x_{ijt}^k \leq u_{ij}^k \left(\alpha_{\bar{i}}^{\bar{k}} + \beta_{\bar{i}\bar{t}}^{\bar{k}} \right) \quad \forall (i, j) \in A^k, \forall k, \bar{k} \in K, \forall \bar{i} \in N'^{\bar{k}} | ((i, k), (\bar{i}, \bar{k})) \in \Psi \text{ or } ((j, k), (\bar{i}, \bar{k})) \in \Psi, \forall t \in T. \quad (54)$$

$$y_{ij}^k \in \{0, 1\} \quad \forall (i, j) \in A'^k, \forall k \in K \quad (55)$$

$$y_i^k \in \{0, 1\} \quad \forall i \in N'^k, \forall k \in K \quad (56)$$

$$z_{ij}^k \in \{0, 1\} \quad \forall (i, j) \in A'^k, \forall k \in K \quad (57)$$

$$z_i^k \in \{0, 1\} \quad \forall i \in N'^k, \forall k \in K \quad (58)$$

$$m_{it}^k \geq 0 \quad \forall i \in N_-^k, \forall k \in K, \forall t \in T \quad (59)$$

$$f_i^k \in \{0, 1\} \quad \forall i \in N'^k, \forall k \in K \quad (60)$$

$$\alpha_{ij}^k \in \{0, 1\} \quad \forall (i, j) \in A'^k, \forall k \in K \quad (61)$$

$$\alpha_i^k \in \{0, 1\} \quad \forall i \in N'^k, \forall k \in K \quad (62)$$

$$\alpha_{ijt}^{kr} \in \{0, 1\} \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T, \forall r \in R^k \quad (63)$$

$$\alpha_{it}^{kr} \in \{0, 1\} \quad \forall i \in N'^k, \forall k \in K, \forall t \in T, \forall r \in R^k \quad (64)$$

$$\beta_{ijt}^k \in \{0, 1\} \quad \forall (i, j) \in A'^k, \forall k \in K, \forall t \in T \quad (65)$$

$$\beta_{it}^k \in \{0, 1\} \quad \forall i \in N'^k, \forall k \in K, \forall t \in T \quad (66)$$

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Conflict of interests The authors have no competing interests to declare that are relevant to the content of this article.

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